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<https://knb.ecoinformatics.org/data/wri-data-processing/cogs/>

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I. Wildfire Resilience Index Framework.

Table 1. Equation symbols

Symbol	Meaning
WRI	Wildfire resilience index
D	domain
R	resilience
S	status
T	resistance
P	recovery potential
I	indicator
C/c	Current value
G/g	reference

To estimate the Wildfire Resilience Index (WRI), we quantify the status (i.e., current condition) and fire resilience of eight domains. The domains capture widely-held public values that landscapes provide to people, and include Infrastructure, Communities, Livelihoods, Sense of Place, Species, Habitats, Water, and Air. By spanning both social and ecological dimensions, the Index reflects the reality that people are embedded in landscapes, and that the factors shaping resilience vary across domains. The WRI is calculated as the average of the domain (D) score for all i domains:

$$WRI = \frac{1}{n} \sum_{i=1}^n D_i \quad (\text{Eq. S1})$$

Each domain score is calculated as the average of the domain status (S) and its resilience to wildfire (R), such that:

$$D_i = \frac{S_i + R_i}{2} \quad (\text{Eq. S2})$$

We define status as the current condition (C) compared to its reference or goal (G)

$$S_i = \frac{C_i}{G_i} \quad (\text{Eq. S3})$$

and resilience as the inverse product of resistance (T) and recovery potential (P):

$$R_i = 1 - (1 - T_i) \times (1 - P_i) \quad (\text{Eq. S4})$$

Resilience is formulated such that values can be high if either resistance or recovery are high; meeting both conditions is not necessary.

Status, Resistance, and Recovery are typically calculated from more than one indicator and are thus a composite sub-index of the WRI.

Rescaling indicators

Every indicator is rescaled from zero to one, where one represents a high score. The value of each indicator (I) is the current value (c) compared to a reference point (g) such that:

$$I_i = \frac{c_i}{g_i} \quad (\text{Eq. S5})$$

We identified a reference point for each indicator; references are determined based on the following five criteria, depending on the goal and data limitations (see Table 2). First, reference conditions can be estimated mechanistically through a **functional** response using a production or growth function (e.g., carrying capacity). Second, references can be derived through **comparison** across geographies with references set by maximal or ideal values (e.g., hospital access). Third, reference conditions can be anchored, either temporally to a **historical** benchmark (e.g., historical habitat extent). Fourth, references can be set to a known or established baseline or threshold relevant to health or **policy** (e.g. AQL, poverty). Finally, references can be a **moving window**, tracking relative change, or lack of change, over a shorter period of time. Although no indicators currently use this moving window approach, it remains a conceptual option. We emphasize the Index is not a predictive framework and should not be interpreted as wildfire risk or hazard. While our indicators focus on mechanistic drivers of resilience, they are indicators, not models.

II. Design Criteria for Indicator Development and Data Selection.

To ensure the Index captures key dimensions of wildfire resilience, indicators were selected based on the following ten criteria:

Scientific Rigor

1. **Defensible and evidence-based:** Indicators must reflect the best available science and have a clearly defined reference point or benchmark.
2. **Non-redundant and complementary:** Each indicator should contribute unique information, complementing rather than duplicating existing metrics.
3. **Flexible in calculation:** Indicators must support multiple calculation approaches to accommodate differences in data quality, resolution, or availability.

Responsiveness

4. **Sensitive to change:** Indicators must be responsive to shifts in condition, capturing meaningful ecological or social change over time.
5. **Predictable in behavior:** Indicators should respond in a consistent direction and magnitude, enabling valid comparisons across geographies and time.
6. **Policy-relevant:** Indicators must reflect conditions that can be influenced by policy or management actions.

Practicality

7. **Scalable across contexts:** The Index must work across spatial and temporal scales to support diverse applications.
8. **Feasible and cost-effective:** Indicators must rely on data that are readily available, affordable, and practical to update.
9. **Robust to missing data:** Components must tolerate data gaps and uneven quality without compromising interpretability.

Communication & Transparency

10. **Understandable and open:** Indicators must be clearly defined, easy to interpret, and built using open science principles (e.g., version control, open-source code, and transparent assumptions).

III. Weighting Resilience Components for Calculating the Index

Grounded in socio-ecological theory, we partition resilience into two components: resistance and recovery, either of which can confer resilience to a system (Holling 1973; Folke 2006).

However, people may prefer landscapes that are highly resistant, minimizing the need for recovery. In such cases, resilience evaluation could justifiably place greater weight on resistance.

We make both resilience components as well as all indicators informing each component, available on the website for download should users wish to recalculate scores with their own weighting system. We also make these data viewable on the dashboard.

IV. Data Inclusion and Data Gaps

Because our application of the Index covers a range of domains across two countries (USA and Canada), data sources are often at different spatial and temporal resolutions and sometimes differ between countries. To accurately compare across geographies, we required that each data source cover the entire study region. Missing data would bias Index scores and reduce comparability. Where possible, we addressed data gaps by merging different datasets into unified layers or by modeling missing values. We used only existing, publicly available datasets to enable open, reproducible, and freely accessible recalculation of the Index. Accordingly, we prioritized wall-to-wall datasets that are regularly updated and have strong external support or legislated mandates.

Notably, the Index is designed as a flexible framework that can incorporate improved datasets as they become available.

V. Geospatial Calculations

Spatial Harmonization. The index incorporates information that is natively at different spatial and temporal resolution (Table 2). To enable integration, we harmonized all layers to a common spatial resolution of 90 meters. Depending on the source, this required either downscaling coarse-resolution data or aggregating finer-resolution layers to match this target resolution.

Interpolation of Point Data. In many cases, data were represented as points and needed to be converted into continuous geospatial layers. To create these geospatial indicators, we spatially interpolated values using kernel density estimates (KDE) and inverse distance weighting (IDW). KDE estimates density or access by calculating point density relative to proximity within the area. This method involves setting a bandwidth (sigma) or radius around each point (e.g., a hospital), defining the distance over which it influences surrounding locations. Cells closer to a point receive higher KDE values, while those farther away receive lower values. IDW operates on the principle that the influence of a point diminishes with distance and more heavily weights the nearer point in the interpolation calculation. We applied KDE to estimate hospital access and fire resource density, while IDW was used for air sensor data, following EPA recommendations.

Distance and Area Calculations. For many indicators, we also calculated distance or area, including interpolating points into continuous layers. In all cases, we used a Mollweide

projection, an equal-area projection, to prevent distortion of area and distance measurements. This approach improves the accuracy of spatial analyses and comparisons across regions, especially at continental scales.

Census Data. Many of our data sources are from the Census (both US and Canadian). The US census provides information in a hierarchical structure, where the census block is the smallest unit, then block groups (600-3,000 people), census tract (1,200-8,00 people), county, and state. The Canadian Census also has a hierarchical structure, where the census block (or block face) is the smallest unit, followed by census dissemination areas (DAs) (similar to U.S. block groups, with 400-700 people), census tracts (CTs) (similar to U.S. census tracts, with about 1,200-8,000 people), census subdivisions (CSDs) (similar to U.S. counties), and provinces or territories. The trade-off is more granular data carries higher uncertainty. Thus, to balance our need for both high accuracy and high resolution, we focus on census tract, which captures a large enough sample size to be representative of a community. When not available, we move to the next finest resolution. In the U.S., we primarily use the American Community Survey (ACS) 5-year estimates for census tracts and smaller units. The ACS collects detailed demographic, social, and economic information annually, but 5-year aggregated estimates are used to improve reliability at fine spatial scales.

Domain Masking. Some domains apply only to human-inhabited areas (Air, Infrastructure, Communities, Livelihoods), while others pertain solely to natural landscapes (Natural Habitats). For these, we masked out either human or natural areas accordingly. When aggregating to larger geographies, masked values were excluded from calculations. Other domains span natural and built systems and are not masked (Water, Species, Sense of Place). We provide both masked and unmasked indicator layers for use.

Gapfilling Methods. To address missing values that primarily arose when vectors were not perfectly aligned, we applied gapfilling procedures to fewer than 0.5% of all cells in the dataset. Two complementary approaches were used at the final stage of processing: vector-based and raster-based gapfilling.

Vector-based gapfilling was applied in three primary contexts: (i) gaps along the outer fringe of the study area, (ii) gaps located between adjacent shapefiles, and (iii) small areas of missing data that were not addressed by the hierarchical backfilling procedure (e.g., census tracts, national boundaries, or ecoregions). In this approach, unfilled gaps were first converted into points, which were then assigned to the nearest vector shape based on Euclidean distance. Each point was subsequently attributed with the value of the associated vector polygon.

Raster-based gapfilling consisted of two sequential steps. First, a cascade aggregation procedure was used: unfilled cells were identified, and values were estimated by progressively aggregating the indicator to coarser spatial resolutions and then disaggregating to fill missing values. Second, remaining gaps were filled using a nearest-neighbor approach, implemented through a Voronoi tessellation-based assignment.

VI. Assumptions and Caveats

Our Index calculation requires two key assumptions. First, we assumed that domains and their indicators do not interact across goals in ways beyond those that are explicitly built into our calculation. This assumption allows us to evaluate each domain independently, without needing to model complex feedback loops across domains. Second, we assumed that the recovery indicators reflect recovery potential, not recovery speed (e.g., engineering resilience, Holling [1996](#)). Systems with naturally long recovery trajectories (e.g., forests) will not be penalized for the length of time required. However, we do acknowledge that recovery rate can interact with disturbance frequency to erode resilience.

VII. Software

All data processing, analysis, and visualizations were performed in R and Python.

VIII. Models for each Goal/Subgoal and indicators

Overview

The WRI is a composite index– the average score of eight socio-ecological domains. For each of our eight socio-ecological domains, we evaluate status and resilience, which are averaged to calculate the domain score. Resilience is calculated as an inverse multiplicative of resistance and recovery (Figure S1). While this is the general framework we used to construct each domain, not all elements were relevant to all domains (Table 2). In some cases, we did not evaluate status because the reference or goal for the domain are too variable and individual. In other cases, we did not evaluate recovery because there either is not sufficient scientific evidence to support recovery indicators or because recovery is on a time scale that is not relevant to policy.

Table 2. List of domains and elements included.

Domain	Status	Resilience	Resistance	Recovery
Infrastructure	N	Y	Y	Y
Communities	N	Y	Y	Y
Livelihoods	Y	Y	Y	Y
Sense of Place: Iconic places	N	Y	Y	Y
Sense of Place: Iconic species	Y	Y	Y	Y
Species	Y	Y	Y	Y
Habitats	Y	Y	Y	Y
Water	Y	Y	Y	N
Air	Y	Y	Y	N

Infrastructure

OVERVIEW

People care about infrastructure, particularly buildings, because they provide the foundation for communities to live, work, and interact. Buildings offer shelter, safety, and access to resources, while also being key to economic development and social stability. While structure loss is one of the most obvious costs of wildfires in the U.S., loss and associated direct costs are not tracked in a standardized way. However, we do have case studies. For example, early estimates over \$59 billion in property and loss from the January 2025 Los Angeles fires, with associated insurance losses ranging between \$20 and \$45B (Li and Yu [2025](#)).

Where buildings are present, this domain assesses their resilience to wildfire. Resistance is characterized through indicators that suggest that the structure is less likely to burn: structural and regulatory features, such as building codes; site-level defensibility factors, including WUI designation, defensible space, and access routes; and systemic protection, represented by local firefighting resource density. Recovery, in contrast, captures community-level characteristics that shape the ability to materially recover after structure loss. These include financial capacity (e.g., median wealth), governance capacity (e.g., whether an area is incorporated), and housing stability (e.g., the proportion of homeowners), all of which contribute to the pace and extent of post-fire rebuilding and restoration.

MODEL

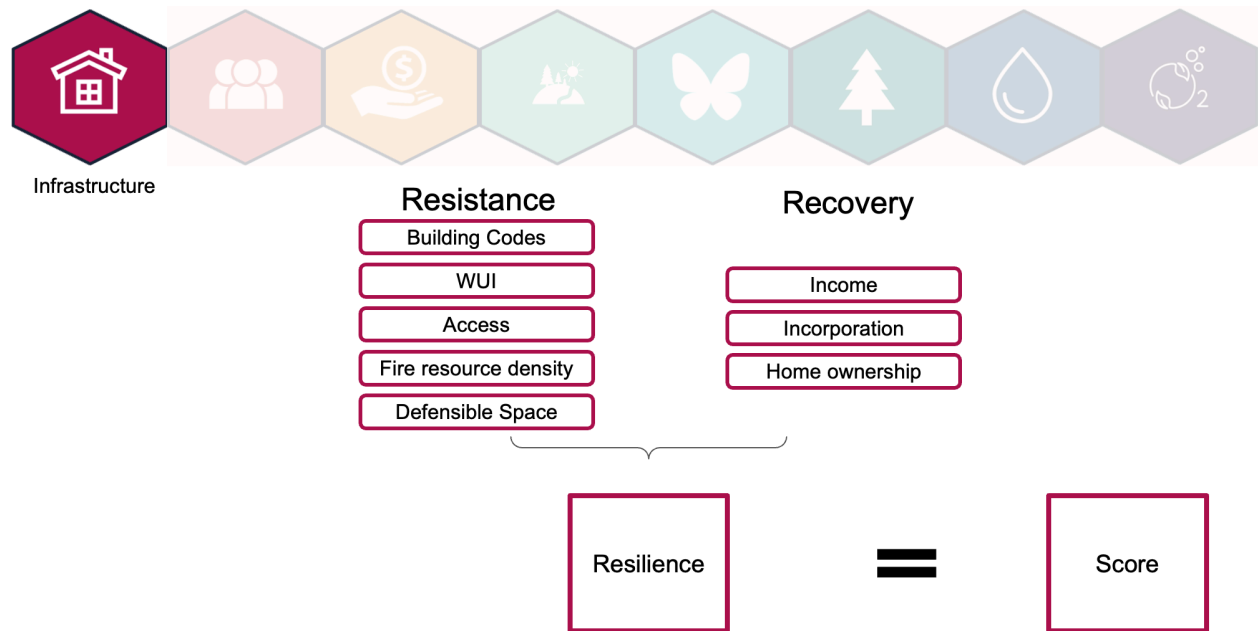


Figure S1. Model for Infrastructure domain.

Presence

This domain does not have a status because why people choose to live where they live is multivariate, and establishing a reference is highly variable based on individual values; in contrast, we are comfortable with the argument that people generally want their infrastructure to be resilient to wildfire. Instead, scores are evaluated where there are structures. We identified structures using [Microsoft Global ML Building Footprint](#). For infrastructure, we consider all structures, spanning commercial and residential, which receive equal weighting. We do not consider shared infrastructure (e.g., water, transportation, etc) because these dimensions are generally either included in this domain (transportation) or others (water). This domain is masked to the [Global Human Settlement Layer](#) (Pesaresi et al. [2024](#)) so that Infrastructure is only evaluated where there are people.

Resilience

Resilience indicators are those that indicate reduced probability of burning and enhanced recovery of structures after an event.

Resistance

Building codes. Building codes play a crucial role in protecting communities by establishing standards for the design and construction of buildings to ensure safety, durability, and resilience to hazards. The International Wildland-Urban Interface Code (IWUIC), developed by the International Code Council (ICC), specifically addresses the unique risks posed by wildfires in

areas where human development meets wildland vegetation (IWUIC [2018](#)). It is used by local governments throughout the United States and internationally to enforce defensible space, promote the use of fire-resistant materials, and implement community-wide fire mitigation practices. The IWUIC sets minimum requirements for planning, construction, and maintenance to reduce wildfire vulnerability in these high-risk zones. It organizes requirements based on severity of risk, with Class I associated with highest risk areas; we assess these Class I requirements. This code is increasingly important as expanding development into fire-prone landscapes elevates wildfire risks to lives, property, and natural resources, making safer building and land management practices essential for community resilience.

In the US, Executive Order 13728, issued on May 18, 2016, mandates that all new federal buildings exceeding 5,000 gross square feet, located on federal land within the WUI and at moderate or greater wildfire risk, must comply with the most recent edition of the IWUIC (EO [13728](#)). Furthermore, the Federal Emergency Management Agency (FEMA) enforces compliance with the IWUIC for public facilities in the WUI as a condition for receiving federal assistance under the Public Assistance Program (FEMA [2021](#)). Specifically, FEMA's policy requires that the repair, restoration, reconstruction, or replacement of public facilities damaged or destroyed by a major disaster must conform to the latest published editions of relevant consensus-based codes, specifications, and standards that incorporate hazard-resistant design. In addition to U.S. federal policy alignment with the IWUIC, the IWUIC's international relevance provides a consistent baseline for evaluation across the U.S. and Canada.

We developed a composite sub-index to evaluate the presence and strength of wildfire-related building codes at the state, territory, or provincial level, scoring both whether a jurisdiction had building codes and the inclusion of specific sections of the IWUIC (Table 3). This approach establishes a baseline for each jurisdiction, but there are two caveats that may cause actual codes in practice to diverge from this baseline. First, building codes typically apply only to new construction, meaning much of the existing housing stock predates these standards and may remain vulnerable. Second, many counties and incorporated areas have adopted stricter local standards, which can raise the effective level of protection above the state baseline.

States received a point if they had a functional statewide building code, formal guidance, a code board, or an equivalent regulatory mechanism. We then scored specific elements of building codes against the IWUIC; states received a single point for each element they had in their code

Table 3. Building code scores.

Jurisdictional Information on WUI Building Codes	Jurisdiction	Alaska	Arizona	California	Colorado	Idaho	Montana	Nevada	New Mexico	Oregon	Utah	Washington	Wyoming	British Columbia	Yukon
	Statewide Wildfire Code	FALSE	FALSE	TRUE	FALSE	FALSE	FALSE	TRUE	FALSE	TRUE	TRUE	FALSE	FALSE	FALSE	FALSE
	Statewide Guidance	FALSE	FALSE	FALSE	FALSE	FALSE	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	TRUE	FALSE	FALSE
	Statewide Code Board	FALSE	FALSE	FALSE	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE
	Something in the Works	FALSE	FALSE		TRUE	TRUE	FALSE	TRUE	TRUE	FALSE	FALSE	TRUE	FALSE	FALSE	FALSE
	Applies to What?			New buildings and remodels (new construction) in the WUI or any Fire Hazard Severity Zone Since July 1, 2008			Buildings in the WUI								
	When?						Since October 1, 2008								
Section 602 Autism and Special System and Section 604 Maintenance of Driveway Space	Roof Covering (504.2)			FALSE				FALSE		TRUE	TRUE				
	Roof Valleys (504.2.1)			TRUE				FALSE		FALSE	FALSE				
	Protection of Eaves (504.3)			FALSE				FALSE		TRUE	TRUE				
	Gutters and Downspouts (504.4)			TRUE				FALSE		TRUE	TRUE				
	Exterior Walls (504.5)			TRUE				FALSE		TRUE	TRUE				
	Underfloor Enclosure (504.6)			TRUE				FALSE		FALSE	TRUE				
	Appendages and Projections / "unenclosed accessory structure" (504.7)			TRUE				FALSE		TRUE	TRUE				
	Underfloor Areas (of the unenclosed accessory structure) (504.7.1)			TRUE				FALSE		TRUE	FALSE				
	Exterior Glazing / Windows (504.8)			TRUE				FALSE		FALSE	TRUE				
	Exterior Doors (504.9)			TRUE				FALSE		FALSE	TRUE				
	Vents (504.10)			TRUE				FALSE		TRUE	TRUE				
	Vent Locations (504.10.1)			TRUE				FALSE		FALSE	FALSE				
	Detached Accessory Structures (504.11)			TRUE				FALSE		FALSE	TRUE				
	Underfloor Areas (of detached accessory structures) (504.11.1)			TRUE				FALSE		FALSE	FALSE				
	General (602.1) → Sprinklers required in all occupancies in new buildings to meet Class I requirements			TRUE				FALSE		FALSE	FALSE				
	Fuel Modification (603.2)			TRUE				TRUE		TRUE	TRUE				
	Responsible Party (603.2.1)			TRUE				TRUE		TRUE	FALSE				
	Trees (603.2.2) & 604.4)			TRUE				TRUE		TRUE	FALSE				
	Groundcover (603.2.3)			FALSE				TRUE		TRUE	FALSE				
	Modified Area (keep area cleared for fire suppression activities) (604.2)			FALSE				TRUE		FALSE	TRUE				
	Responsibility (604.3)			TRUE				TRUE		TRUE	TRUE				
	Chimney Clearance (related to trees) (604.4.1)			TRUE				TRUE		TRUE	FALSE				
	Deadwood removal (from trees) (604.4.2)			TRUE				TRUE		TRUE	FALSE				

From Section 504: Class I Ignition-Resistant Construction we considered 14 elements: roof covering (504.2), roof valleys (504.2.1), protection of eaves (504.3), gutters and downspouts (504.4), exterior walls (504.5), underfloor enclosure (504.6), appendages and projections or unenclosed accessory structures (504.7, 504.7.1), exterior glazing/windows (504.8), exterior doors (504.9), vents and vent locations (504.10, 504.10.1), and detached accessory structures and their underfloor areas (504.11, 504.11.1).

From Section 602 (General) we scored the requirement for sprinklers in all occupancies in new buildings to meet Class I standards (602.1).

From Section 603: Defensible Space we scored 4 elements: fuel modification (603.2), responsible party designation (603.2.1), tree management (603.2.2), and groundcover maintenance (603.2.3).

From Section 604: Maintenance of Defensible Space we scored four elements: maintaining cleared areas for fire suppression activities (604.2), responsibility for maintenance (604.3), chimney clearance related to trees (604.4.1), and deadwood removal (604.4.2).

To convert these scores into indicators, we summed the number of points and rescaled such that zero elements resulted in a resistance score of zero and inclusion of all possible elements resulted in a score of one; intermediate values were linearly interpolated.

Defensible space. Defensible space is a critical component of wildfire resistance, and refers to the strategically managed area between a structure and surrounding vegetation that reduces fire intensity and slows its spread. Creating defensible space is often described as one of the most important actions an individual homeowner can take to increase the resistance of their home to wildfire. Three potential explanations exist for how defensible space could help reduce structure loss. First, defensible space creates a buffer around a structure that may prevent wildfire from moving from vegetation to the home, though some have argued for the protective effect is stronger when the vegetation is not water stressed. Second, defensible space may serve as a protective buffer for firefighting personnel, a less hazardous space where they can safely defend a structure. Third, defensible space may act at a community level, limiting the spread of fire between structures. These three scenarios are reviewed in Gill ([2005](#)), Gill and Stevens ([2009](#)), and Syphard et al. ([2012](#)).

To estimate defensible space around structures, we combined two spatial datasets: ESRI's 2023 land cover data and Microsoft's building footprint dataset. Vintage of extracted building footprints depends on the vintage of the underlying imagery. The underlying imagery is from Bing Maps including Maxar and Airbus between 2012 and 2021. We focused on tree cover as a proxy for vegetation that poses a wildfire risk. For each structure, we calculated the percentage of area in a 10 m buffer (Zone 1) that was covered by trees. To convert this into an indicator, we applied a threshold-based linear rescaling: areas with 50% or more tree cover received a score of 1, representing minimal defensible space, while areas with no tree cover received a score of 0, representing optimal defensible space. This threshold value was selected

based on preliminary analysis of structure loss in the Woolsey fire (*unpublished results*). Intermediate values were interpolated linearly between these endpoints.

These scores are at the structure/parcel level, which can be decoupled from fire behavior. To reflect a 'neighborhood' of resistance, we rasterized these structural values to 90m and in aggregating up, we assigned the pixel the lowest defensible space indicator score to reflect evidence that defensible space acts at a community level.

Notably, our measure of defensible space focused only on surrounding tree cover and does not account for other features that can compromise defensible space, such as decks, fencing, propane tanks, or wood piles.

Access and egress. Road infrastructure is key to wildfire resilience, impacting firefighter access to slow or stop incursion of fire and evacuation of local communities (Maranghides and Link [2025](#)). When communities are designed with limited or poor egress options, fire suppression crews may face significant delays or be entirely unable to reach threatened structures. Narrow roads, single access points, or heavy congestion can also impede evacuation.

We quantify vulnerability as the number of access routes. To generate these values, we need to define the boundaries of communities and count access points in/out of those boundaries.

In the United States, we identify communities as Census Places (CPs), which are geographic areas designated by the US Census Bureau to capture spatially defined communities. Shapefiles for these communities were sourced from the US Census Bureau's TIGER/Line repository using a custom Python pipeline on February 13, 2025. In Canada, we identified communities through Canadian Population Centers (PCs) which have a population of at least 1,000 and a population density of 400 persons or more per square kilometer ([ref](#)). To simplify computation we filtered Census Place and Population Centers to those with a population below 50,000, to focus on smaller and potentially more isolated communities where suppression and other resources may be limited. This threshold comes from the US Census Bureau definition of a small town, which includes communities with populations 50,000 ([ref](#)). We focus on these small communities because they are less likely to be embedded in an urban transportation network, and are thus more likely to be vulnerable. For CP's and PC's with a population greater than 50,000, we assumed sufficient access routes and rescaled those to a value of 1. For structures outside of CPs and PCs we assume limited access rounds and rescale those to a value of zero.

We used Open Street Maps (OSM) to count the number of roads that intersect with the boundary of a CP and PC. OSM is a collaborative, user-driven mapping platform that provides freely available, editable geographic data, including roads. We only consider the higher order types of roads (motorway, primary, secondary and tertiary roads), excluding small residential and unclassified roads (e.g., dirt, cycling routes) as we reason these are unlikely to be used for evacuation or bringing resources into a region.

At the highest level, OSM classifies motorways as roads designed for fast, long-distance travel with controlled access and no intersections. Next are trunk roads, which are major routes that

connect cities and regions but have either intersections or lower capacity than motorways. Primary roads are routes between towns and cities, while secondary and tertiary roads form regional and local connections, often linking smaller communities. OSM data was accessed July 11, 2025.

To capture the roads and lanes that intersect with a CP and PC, we first query OpenStreetMap with each for each place expanded by six kilometers; this broad buffer is used solely for downloading candidate roads. We then clip the resulting geometries to a 500 m buffer around the original CP boundary and use that zone to measure ingress and egress. We counted the number of higher order roads that intersected with this boundary to identify where access is more likely to be cut off. We divide this count by two because two lane roads are counted as one in each direction, but functionally are only a single point of entry or exit.

Our recent work shows the number of roads out of a community is a strong predictor of wildfire fatalities, with a critical threshold of six roads (Fong et al. *in press*). To convert the number of access routes into an indicator between zero and one, we rescaled the values such that the lower threshold was zero and the upper threshold was more than six roads. Indicator values were binary. Areas that were not in a CP or PC received a score zero (low resistance).

WUI. Most structures lost to wildfire are located in the Wildland-Urban Interface (WUI)—the zone where human development meets or intermingles with wildland vegetation (Caggiano et al. [2020](#)). Because this area concentrates both ignition risks and structural exposure, the presence and extent of WUI development is a strong indicator of structural vulnerability, making it a meaningful component of a resistance assessment.

To identify the WUI, we used the map created by the [Silvis Lab](#) in collaboration with USDA Forest Service, which defines the existing WUI definition published in the [Federal Register \(USDA and USDI 2001\)](#). This data product classifies land based on vegetation type and housing densities. The classification schema distinguishes between intermix and interface WUI areas, those areas where housing is intermingled with or adjacent to forest, shrubland, wetland, or grassland vegetation. Areas outside the WUI can include areas with no housing or low density housing, densely developed urban or commercial areas, or other non-vegetated areas such as agricultural areas.

To convert the Silvis Lab's WUI categories into a structural resistance indicator, we assigned values that reflect the relative resistance associated with each land class (Table 4). Areas classified as forest- or grassland-dominated intermix or interface WUI received the lowest indicator value (0), reflecting their proximity to flammable vegetation and the mingling of wildland fuels with structures—conditions that elevate structural risk. Intermix and interface WUI received the same score based on evidence that structural losses are similar between the two classes (Kramer et al. [2019](#)). All other land types—including non-WUI vegetated areas, urban zones, and non-vegetated lands—were assigned the highest resistance score (1), reflecting their lower likelihood of structure loss from wildfire. This approach allowed us to capture relative structural risk across the landscape in a way that is both consistent with empirical loss patterns and grounded in land classification data.

Table 4. Lookup table between Silvris Lab code, land type, WUI type, vegetation, and our resistance indicator score.

Code	Land Classification	WUI Type	Dominant Vegetation	Resistance Score
0	Non-Vegetated / Non-WUI	Non-WUI	None	1
1	Intermix WUI	WUI - Intermix	Forest / Shrubland / Wetland	0
2	Interface WUI	WUI - Interface	Forest / Shrubland / Wetland	0
3	Intermix WUI	WUI - Intermix	Grassland	0
4	Interface WUI	WUI - Interface	Grassland	0
5	Non-WUI	Non-WUI	Forest / Shrubland / Wetland	1
6	Non-WUI	Non-WUI	Grassland	1
7	Urban	Non-WUI	Urban	1
8	Other (non-vegetated, uncategorized)	Non-WUI	Other	1

Systemic Protection:

Firefighting Resource Density. Active firefighting is an important driver of structure survival in a wildfire (Syphard and Keeley [2019](#)). Fire resource density reflects the local capacity for emergency response and fire suppression. Areas with a higher density of fire stations—especially those equipped and trained to respond to wildfires—may be more likely to detect, contain, and manage fires promptly, reducing structural losses and enhancing community safety. This indicator focuses on structural firefighting resources and does not capture wildland firefighting capacity.

We use fire stations as a proxy for firefighting resources. Data for the U.S. were sourced from the [Homeland Infrastructure Foundation-Level Data \(HIFLD\)](#) initiative, managed by the U.S. Department of Homeland Security, which provides a comprehensive dataset of fire station locations nationwide. Data for British Columbia were obtained from GeoBC Branch (GeoBC [2025](#)), and for Yukon, from the Wildland Fire Management Organization, Yukon government.

To quantify firefighting resource density, we calculated a kernel density estimate (KDE) of fire station locations, producing a continuous surface that represents relative fire resource availability across space. This spatially explicit approach captures resource distribution more realistically than administrative boundaries, which may not align with actual fire response areas. Regions with higher KDE values indicate a greater concentration of fire protection infrastructure and thus received higher scores on the fire resource density indicator. These values were rescaled between zero and one, where zero represents the lowest density (areas far from firefighting resources) and one represents the highest density (areas near concentrated fire fighting capacity).

Recovery

Community-level indicators

Median Income: Median income reflects the overall earning capacity of a region's residents. It serves as a proxy for economic well-being and the ability to meet basic needs. Income levels often vary with place, influenced by the cost of living, availability of quality jobs, and local economic development. Using median instead of mean downweights extreme outliers on the high income side of the spectrum.

The US reports household median income in USD as its own statistic at the census tract level. US data is retrieved using the R Census API. Canada reports individual median income in CAD as its own statistic at the census subdivision level. This data is downloaded from the Statistics Canada website (Statistics Canada [2025](#)).

To convert median income into a policy-relevant indicator, we applied a relative rescaling approach within each country. The reporting unit with the highest median income was assigned a score of one, and the unit with the lowest a score of zero, with all other values linearly interpolated between these two extremes. Rescaling was conducted separately for the United States and Canada to account for structural differences in social policy—particularly in areas such as healthcare and income supports—which influence both the real value of income and its role in supporting livelihoods.

Incorporation. This data layer describes whether or not a place is legally incorporated under state law. Incorporation is the formalization of a place, allowing it to have its own governmental systems. Incorporated places may have a better ability to obtain aid and direct resources and services important in rebuilding, enabled by local governance structures. An incorporated place may also have a stronger sense of identity through this formalization. The US dataset is from the Census retrieved with *tidycensus*. The Canada dataset is from Canada Statistics 2021, retrieved with the *cancensus*. US data is vector shapes delineating Census Designated Places (CDP) level. Canadian data is vector shapes at the census subdivision level. Census boundaries for each country get updated periodically.

The datasets are filtered to identify incorporated places (US [codes](#) C1, C7, and C8; CAN [codes](#) RGM, CY, DM, VL, and T). We rasterize these vector shapefiles to 100 meters; one is applied to the incorporated place (high recovery) and zero to everything else (low recovery score).

Home ownership. This data layer represents the percent of households that own their home. Home owners may qualify for more rebuilding resources (state and federal aid) than renters. We include mobile homes in this category of homeowners, since owners will also qualify for aid if owned. Renters may be more vulnerable to displacement after disasters, and do not have home equity to draw upon in recovery. Further, homeownership may deepen ties and commitments to community while renters are more labile. US data are from the ACS, retrieved with *tidycensus*. Canadian data are from Canada Statistics 2023 [data](#), downloaded as a csv. Data are at the census tract (US) and subdivision (CAN) level. Data on homeownership is available over five year moving time periods at smaller geographies, through the American Community Survey, while a new updated data for Canada is occasionally.

The US ACS provides the percentage of owner occupied housing units. Canada Statistics provides the number of private households that are renters and owners. Percent for Canada is the owners divided by all private households. We gapfill any missing values with values from the county (US) or division (CAN) level. We combine these datasets and then rescale the data from zero to one where zero is the lowest percent of homeowners (thought to contribute to lower recovery) and one is the highest percent of homeowners (thought to contribute to higher recovery).

Communities

OVERVIEW

Humans are inherently place-based, and characteristics related to community can impact resilience to wildfires. Here, we define communities as a social unit situated within a geographic area which results in shared individual and collective characteristics such as norms, values, customs, and identity. Some individual and collective characteristics can increase community resilience to wildfires.

This domain assesses community resilience where people live. Resilience consists of both individual and collective indicators. These resilience indicators are intended to reflect the potential to minimize loss of life and the potential for a community to persist and/or rebuild in the event of a wildfire; this domain does not capture structure loss or damage, as this is represented in the Infrastructure domain.

MODEL

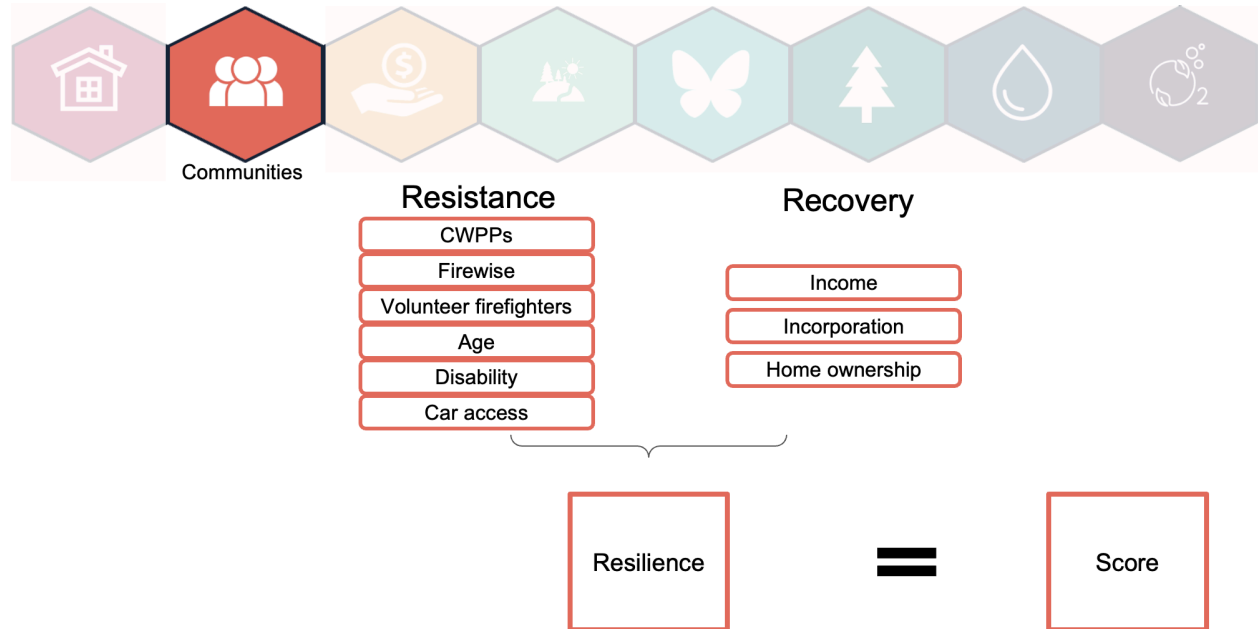


Figure 2. Model for Communities domain.

Presence

Like infrastructure, this domain does not have a status and is only evaluated where there are human communities, identified using the Human Settlement Layer (Pesaresi et al. [2024](#)). We do not evaluate status because where people choose to live is highly subjective and idiosyncratic, and ascribing a value does not contribute insight into the subdomain or the Index as a whole.

The US population data is from the US ACS, retrieved with *tidycensus*. Canadian population data is from Statistics Canada, retrieved with *cancensus*. Population data is at the census tract level (US) and census subdivision level (CAN) and is updated annually and every five years respectively. Communities are masked to where people actually live with the Human Settlement Layer, buffer to 1 km.

Resilience

Resilience encompasses the resistance and recovery of a community to wildfire through individual variables such as demographics as well as collective attributes, such as policy, community agreements, structures, and institutions.

There are six resistance and four recovery indicators. Resistance is calculated by first averaging all the individual and collective indicators separately, and then averaging those two groupings

for a final resistance score. Recovery is calculated the same way, first averaging all the individual and collective indicators and then averaging these individual and collective means.

Resistance

Individual indicators of resistance pertain to the likelihood of an individual surviving a wildfire by evacuating if necessary, and include age (>65), disability, and vehicle access.

Collective level indicators reflect the interest of the community in preparing for and mitigating the impacts of wildfire; these include Community Wildfire Protection Plans (CWPPs), Firewise communities, and volunteer fire departments.

Individual and collective indicators are first averaged separately, then combined for a mean resistance score.

Individual indicators

Percent of population 65+. We calculate the percent of the population 65 or older because older people may face greater challenges with wildfire preparation, mitigation, and evacuation and make up the majority of fatalities during wildfire events ([FEMA](#)). US data are from the ACS, retrieved with *tidycensus*. Canadian data are from Canada Statistics, retrieved with *cancensus*. Data are at the census tract (US) and subdivision (CAN) level; data are updated annually and every five years, respectively.

The US ACS provides percent of the population ≥ 65 . The Canada Census provides population sizes, from which we calculated percent of the population ≥ 65 . We gapfill any missing values with those from county (US) or division (CAN) values. We combined information from both countries and rescaled the data from 0 to 1 such that tracts with the lowest percent of people 65+ had the highest scores. We then inverted the data ($1 - \text{percent } 65^+$) so that tracts with more people 65+ had lower resistance scores.

Disability. We calculate the percent of the population that is disabled because disabled people face additional challenges with wildfire preparation, mitigation, and evacuation (Gaskin et al. [2017](#)). This is defined by the US as *Hearing difficulty, Vision difficulty, Cognitive difficulty, Ambulatory difficulty, Self-care difficulty, Independent living difficulty*. The definition for Canada includes *Hearing, Vision, Mobility, Flexibility, Dexterity, Pain-related, Learning, Developmental, Mental health-related, Memory*. US data are from the ACS, retrieved with *tidycensus*. The Canada dataset comes from the Canadian Survey on Disability (CSD) conducted by Statistics Canada (www150.statcan.gc.ca). Data are at the census tract (US) province level (CAN); data are updated annually and every five years respectively.

Both datasets have the proportion of the population with a disability (US, noninstitutionalized civilians ≥ 5 years old; Canada ≥ 15 years old). We gap filled any missing values with values from the county level for the US; no gap filling was needed for Canada. We combine these datasets and rescale the data from 0 to 1 where 1 represents the smallest fraction of the population living

with a disability. We then invert this indicator (1-disability) so that higher scores represent a higher proportion of people with a disability.

Vehicle access. We calculate the percent of the households with no vehicle access because a lack of vehicle access makes evacuation more difficult (Grajdera et al. [2022](#)). The US data are from the ACS, retrieved with *tidycensus*. There is no equivalent dataset for Canada. Data is at the census tract level and data is updated annually.

The Census reports the proportion of households with no vehicle access. We gapfill any missing values with county estimates. We rescale the data from 0 to 1 such that 1 represents the largest proportion of households with access to a vehicle. We then invert the data (1-vehicle access) so that higher scores represent a higher proportion of households with access to a vehicle.

Collective indicators

All of these collective indicators are only available at the county-level. In practice, these forms of resistance are not distributed evenly across the landscape. As a result, our approach likely overestimates the level of collective resistance in areas where individual or community participation is lower.

Community Wildfire Protection Plans (CWPPs). This data layer describes the number of Community Wildfire Protection Plans (CWPPs). CWPPs are community-driven, collaborative frameworks that identify priorities for mitigating wildfire risk. In the US, the Healthy Forest Restoration Act (HFRA) supports the development of CWPPs. They are a relatively common planning tool for increasing community resilience and strengthening collaboration and coordination across state and federal agencies, local government, and communities. Having a CWPP qualifies communities for some federal grants related to wildfire mitigation. Canada's "equivalent" of CWPPs, the Community Wildfire Resiliency Plan (CWRPs), are kept private at the individual level and are not publicly available; this indicator can not be evaluated in Canada.

For the continental US (CONUS), data come from the Community Wildfire Protection Plan Portal and Data Library Google form. For Alaska, we manually created a dataset following methods similar to those used for the Community Wildfire Protection Plan Portal and Data Library. The CWPP data contains vector shapefiles for locations of each CWPP. The CONUS dataset is updated as new plans are submitted by a team led by Matt Hamilton (hamilton.1323@osu.edu). The dataset we generated for Alaska will be ingested into this portal in the near future.

The vector shapes are rasterized to 90 meters and the sum of how many shapes overlap were calculated for each cell. There can be more than one CWPP in a cell because plans exist at various levels (e.g., county, fire protection district, community). For example, a county could be represented by a plan, but a community within the county may also have its own plan. However, we do not anticipate having plans at multiple jurisdictional levels translates into increased community resistance to wildfire. Thus, the CWPP indicator is then rescaled between zero and one where zero means no CWPPs (lower resistance score) and one means at least one CWPP (higher resistance).

Firewise communities. This data layer describes the number of Firewise communities. A Firewise community meets designated criteria for wildfire preparedness and thereby obtains national recognition ([Firewise USA](#)). This designation signals that the community has worked together on measures to resist structural damage from wildfires. This designation indicates a higher sense of community cohesion and connectedness regarding wildfire. The National Fire Protection Association lists active Firewise communities on their [website](#). Data for FireSmart communities–the Canadian analogue– are privately held and not evaluated.

US Firewise community data contains community information along with county and state information (exact format varies by community). Spatial information in the form of geometries (i.e. vector shapefiles) are not included with this dataset and are also not available by request. It is unclear how often the Firewise community list is updated. The information is only well enough resolved to count the number of Firewise Communities at the county level. Values are then rasterized to a 90 meter grid. The indicator is rescaled between zero and one such that zero represents no Firewise Communities (low resistance) and one represents at least one Firewise communities (higher resistance).

Volunteer fire departments. This data layer describes the number of volunteer fire departments in a county (US) or division (CAN). The motivation to volunteer to fight fires is a good indicator of a community's awareness of wildfire risk and desire to engage in preparation and mitigation efforts. The US dataset was downloaded as a csv from the [USFA](#). The BC dataset was text scraped from a [government pdf](#) into a data table format in R. Volunteer fire departments in the US and BC are resolved to department address level. The Yukon dataset was provided by the [Wildland Fire Management Organization](#) in the Yukon government.

We filter US dataset by fire department type classification, keeping Volunteer and Mostly Volunteer. For the BC dataset, we keep only fire departments with 'volunteer' or 'vol' in the name. We count the number at the county (US) or division (CAN) level and rasterize to 90 meters. The indicator is then rescaled between zero and one such that zero reflects lower numbers of volunteer fire departments (low resistance) and 1 reflects higher number of volunteer fire departments (high resistance).

Recovery

Indicators of recovery pertain to the ability to rebuild a community after a fire. Individually, these include income and home ownership. Collectively, this is incorporation.

Individual indicators will be averaged separately first, then averaged with the single collective indicator for a mean resistance score.

Individual indicators

Home ownership. This data layer represents the percent of households that own their home. Home owners may have more ease with rebuilding–and thus returning to their community–than renters because home ownership qualifies owners to resources (state and federal aid) unavailable to renters. Mobile homes (if they are owned) qualify for this aid and therefore are

not considered separately. Further, homeownership may deepen ties and commitments to community while renters are more labile. US data are from the ACS, retrieved with *tidycensus*. Canadian data are from Canada Statistics 2023 [data](#), downloaded as a csv. Data are at the census tract (US) and subdivision (CAN) level. A new dataset for the US is available every 5 years, while a new updated data for Canada is occasional.

The US ACS provides the percentage of owner occupied housing units. Canada Statistics provides the number of private households that are renters and owners. Percent for Canada is the owners divided by all private households. We gapfill any missing values with values from the county (US) or division (CAN) level. We combine these datasets and then rescale the data from zero to one where zero is the lowest percent of homeowners (low recovery) and one is the highest percent of homeowners (high recovery).

Income – greater than \$200k. This data layer represents the percent of the community with income >\$200k. We include this information because individuals or households with higher income may ease household recovery (i.e. less financial strain in rebuilding). US data are from the ACS, retrieved with *tidycensus*. Canada data are from Canada Statistics 2021 data, retrieved with the *cancensus*. We use data at the census tract (US) or subdivision (CAN) level; new datasets are available every year and every five years respectively.

The US ACS provides the proportion of households earning \$200k or more income in the past 12 months. Canada Statistics provides the number of households with \$200k or more income and total households surveyed for income. From this, we calculate the proportion of households earning >\$200k. We gap fill missing values with values from the county (US) or division (CAN) level. We combine these datasets and then rescale the data from zero to one such that zero represents the lowest proportion of high income earners (low recovery) and one represents the highest proportion of high income earners (high recovery).

Income – below the poverty level. This data layer describes the percent of households below the poverty line. Households living below the poverty line may find it more challenging to recover from disaster (i.e. more financial strain). Disaster financial assistance is often allocated by household without accounting for multigenerational living, which can put lower-income families living together at a disadvantage (Moloney et al. 2023). US data are from the ACS, retrieved with *tidycensus*. Canada data comes from Canada Statistics 2022 [data](#), downloaded as a csv. Data at the census tract (US) or subdivision (CAN) level. Updates for the US are annual; updates for Canada are occasionally.

The US ACS provides the proportion of the population below the poverty level. Canada Statistics provides the number of people in poverty based on individual MBM poverty status (Canada's official measure of poverty), from which we calculate percentages. We gap fill missing values with values from the county (US) or division (CAN). We combine these datasets and rescale the data from zero to one where zero is the fewest people in poverty and one is the most. We then invert the data (1-percent in poverty) so that zero is a higher proportion of the population below the poverty line (lower recovery) and one is a lower proportion of the population below the poverty line (higher recovery).

Collective indicator

Incorporation. This data layer describes whether or not a place is legally incorporated under state law. Incorporation is the formalization of a place, allowing it to have its own governmental systems. Incorporated places may have a better ability to obtain aid and direct resources and services important in rebuilding, enabled by local governance structures. An incorporated place may also have a stronger sense of identity through this formalization. The US dataset is from the Census retrieved with *tidycensus*. The Canada dataset is from Canada Statistics 2021, retrieved with the *cancensus*. US data is vector shapes delineating Census Designated Places (CDP) level. Canadian data is vector shapes at the census subdivision level. Census boundaries for each country get updated periodically.

The datasets are filtered to identify incorporated places (US [codes](#) C1, C7, and C8; CAN [codes](#) RGM, CY, DM, VL, and T). We rasterize these vector shapefiles to 100 meters; one is applied to the incorporated place (high recovery) and zero to everything else (low recovery score).

Livelihoods

OVERVIEW

Livelihoods (jobs) are important because they represent how people make a living. Beyond just income, though, livelihoods are deeply connected to dignity, security, identity, and well-being. The number of jobs in a community, the stability of those jobs and potential for job growth, and the range of job types are all key metrics of community stability, quality of life, and overall local economic opportunities.

We measure livelihoods as the interplay of employment, income, and housing affordability, elements that are of direct relevance to individuals and policymakers. Further, the resilience to wildfire varies across job types, with some jobs (such as in the tourism industry) more vulnerable to wildfires.

MODEL

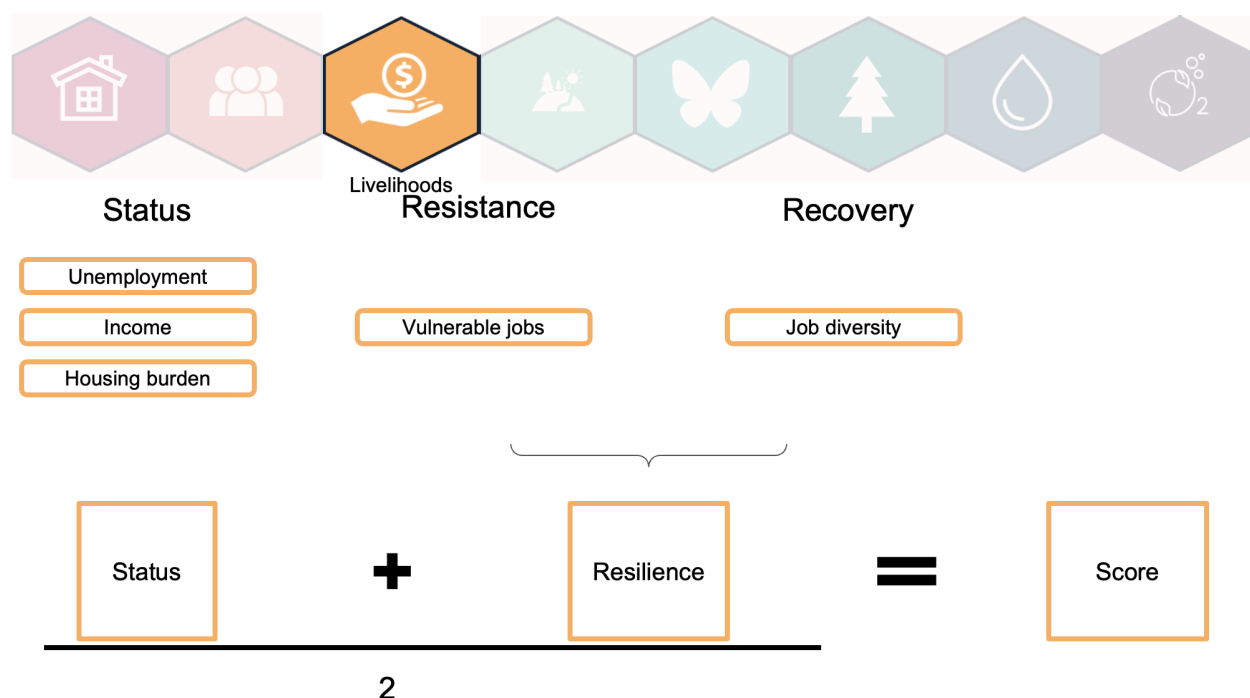


Figure 3. Model for Livelihoods domain.

Status

Unemployment. Employment is a cornerstone of economic stability. High unemployment signals limited job opportunities, which can undermine financial independence and increase vulnerability. The US reports unemployment rate at the census tract level. US data is retrieved using the R tidycensus API. Canada reports unemployment as its own statistic at the census subdivision level. This data is downloaded from the Statistics Canada website.

We normalized unemployment rates to a zero to one scale, where a value of one represents strong labor market performance and zero represents weak performance. Specifically, we set 4% unemployment equal to one and 10% unemployment equal to zero. This choice is supported by economic literature and policy conventions. An unemployment rate near 4% is often considered consistent with “full employment” or the non-accelerating inflation rate of unemployment (NAIRU), reflecting a healthy labor market where job seekers can generally find work without exerting inflationary pressure (Ball & Mankiw [2002](#), Elsby et al. [2010](#), Crump et al. [2019](#)). By contrast, unemployment rates approaching 10% are historically associated with severe recessions and significant labor market distress, such as during the Great Recession (Elsby et al. [2010](#)). Setting these anchors allows the indicator to reflect both the desirable policy target and the threshold for serious economic stress.

Median Income: Median income can serve as a proxy for economic well-being and the ability to meet basic needs. Income levels often vary with place, influenced by the cost of living, availability of quality jobs, and local economic development. Using median instead of mean downweights extreme outliers on the high income side of the spectrum.

The US reports household median income in USD as its own statistic at the census tract level. US data is retrieved using the R tidycensus API. Canada reports individual median income in CAD as its own statistic at the census subdivision level. This data is downloaded from the Statistics Canada website.

To convert median income into a policy-relevant indicator, we applied a relative rescaling approach for tracts within each country. The reporting unit with the highest median income was assigned a score of one, and the unit with the lowest a score of zero, with all other values linearly interpolated between these two extremes. Rescaling was conducted separately for the United States and Canada to account for structural differences in social policy—particularly in areas such as healthcare and income supports—which influence both the real value of income and its role in supporting livelihoods.

Housing Burden. Housing burden measures the share of income spent on housing costs. When households spend a high proportion of their income on rent or mortgages, less remains for essentials like food, healthcare, and education. Housing affordability is deeply tied to place, shaped by zoning, land use, and market pressures.

The US reports their housing statistics by rent burdened housing units for each housing tenure (mortgage status). This is then combined and divided by the total number of housing units to produce the percentage of housing units that are rent burdened, which is defined as paying 30% or more of a household income on housing. Canada reports their housing statistics by number of households paying 30% or more on housing and total number of housing units. This is then used to calculate the percentage of housing units that are rent burdened.

To convert housing burden into an indicator, we used a relative rescaling approach, assigning a score of one to the reporting unit with the lowest housing burden and zero to the unit with the highest burden. Intermediate values were linearly interpolated. This process was conducted separately for the United States and Canada, acknowledging differences in housing markets, cost of living, and public services (such as subsidized healthcare) which shape how much of a household's income is available for housing and other essential needs.

Resilience

Resistance

Job Vulnerability. Wildfires can disrupt local economies by displacing workers, damaging infrastructure, or reducing access to workplaces. However, the degree of disruption to the labor

market is highly uncertain and likely to vary across time and place. Evidence suggests that service industry jobs and jobs that rely heavily on tips and seasonal work may be particularly vulnerable (Davis et al. [2014](#), Kim and Jakus [2019](#), Addoum et al. [2023](#)). Thus, sectors that rely on tourism or in-person services are often more susceptible to disruption during and after a wildfire. By estimating the proportion of jobs in a region that fall into these vulnerable sectors, we can better understand a region's economic exposure to wildfire and its capacity to maintain livelihoods in the face of fire-related disruption. This measure provides insight into both short-term shocks and long-term recovery potential.

We calculate the total number of jobs that are affected by wildfires by first identifying the number of vulnerable workers (Table 5) divided by the total jobs per area to create a percentage of jobs that are highly impacted by wildfire. We use the four digit North American Industry Classification System (NAICS) to identify these jobs comparable between countries.

To tabulate the number of workers in different employment sectors in the US, we use the Quarterly Census of Wages and Employment (QCEW), a comprehensive dataset produced by the U.S. Bureau of Labor Statistics (BLS) in partnership with state workforce agencies. The QCEW collects detailed information on employment and wages from employers who are subject to state unemployment insurance laws. This Census covers 95% of jobs in the US. To tabulate the number of workers in different employment sectors in Canada, we use Statistics Canada, which provides the number of workers for each NAICS code for each district in Canada.

To convert vulnerable jobs into an indicator, we used a relative rescaling approach, assigning a score of one to the reporting unit with the lowest percentage of vulnerable jobs and zero to the unit with the highest percentage of vulnerable jobs. Intermediate values were linearly interpolated. This process was conducted separately for the United States and Canada, acknowledging differences in housing markets, cost of living, and public services, such as subsidized healthcare, which shape how much of a household's income is available for housing and other essential needs.

Table 5. Vulnerable jobs by NAICS code.

NAICS Industry Code	NAICS Code Industry Description
4871	Scenic and sightseeing transportation, land
4872	Scenic and sightseeing transportation, water
4879	Scenic and sightseeing transportation, other
5322	Consumer goods rental
5323	General rental centers
5615	Travel arrangement and reservation services
7112	Spectator sports
7113	Promoters (presenters) of performing arts, sports and similar events
7115	Independent artists, writers and performers
7131	Amusement parks and arcades
7132	Gambling industries
7139	Other amusement and recreation industries
7211	Traveler accommodation
7212	Recreational vehicle (RV) parks and recreational camps
7213	Rooming and boarding houses
7223	Special food services
7224	Drinking places (alcoholic beverages)
7225	Full-service restaurants and limited-service eating places

Recovery

Diversity of Jobs. A diverse mix of jobs and industries enhances a region's ability to recover from wildfire impacts. Wildfires often disproportionately affect certain sectors (e.g., the service industry); when a local economy is not overly dependent on these vulnerable industries, workers can find employment in other sectors less affected by wildfire, helping to stabilize incomes and reduce economic disruption. This economic diversity spreads risk and supports faster, more resilient recovery for communities facing wildfire events.

To assess the diversity of employment opportunities within a region, we calculate the Shannon diversity index using the number of jobs and employees classified by NAICS codes for each area. This index captures both the variety and relative abundance of different industries, reflecting how broadly employment is distributed across economic sectors. A higher diversity score indicates a more varied and potentially resilient local economy, which can better withstand disruptions such as wildfires.

To facilitate comparison across regions and countries, the Shannon diversity index is rescaled to range from one to zero separately within each country such that zero reflects high diversity and one reflects low diversity. This normalization accounts for structural differences in economic composition and reporting between countries, allowing for consistent interpretation of the indicator as a measure of economic diversity and recovery potential.

Sense of Place

People often feel a deep connection to particular places—through lived experience, heritage, or simply knowing these places exist. These connections can be shaped by the presence of distinctive species, community connections, or the character of the landscape itself. This goal captures those connections by assessing how iconic species and places are currently doing, and how they may be affected by wildfire. The condition and resilience of these species and places offer insight into the cultural, spiritual, aesthetic, and other intangible values people associate with fire-prone landscapes, and whether those values are being maintained in the face of growing fire risk.

Iconic Places

OVERVIEW

Iconic places are areas that are culturally significant for a variety of reasons. Designating places as a landmark, monument, park, and/or Protected Area is an indicator that people value those places for their ecosystem services or the cultural heritage and identity they represent.

MODEL

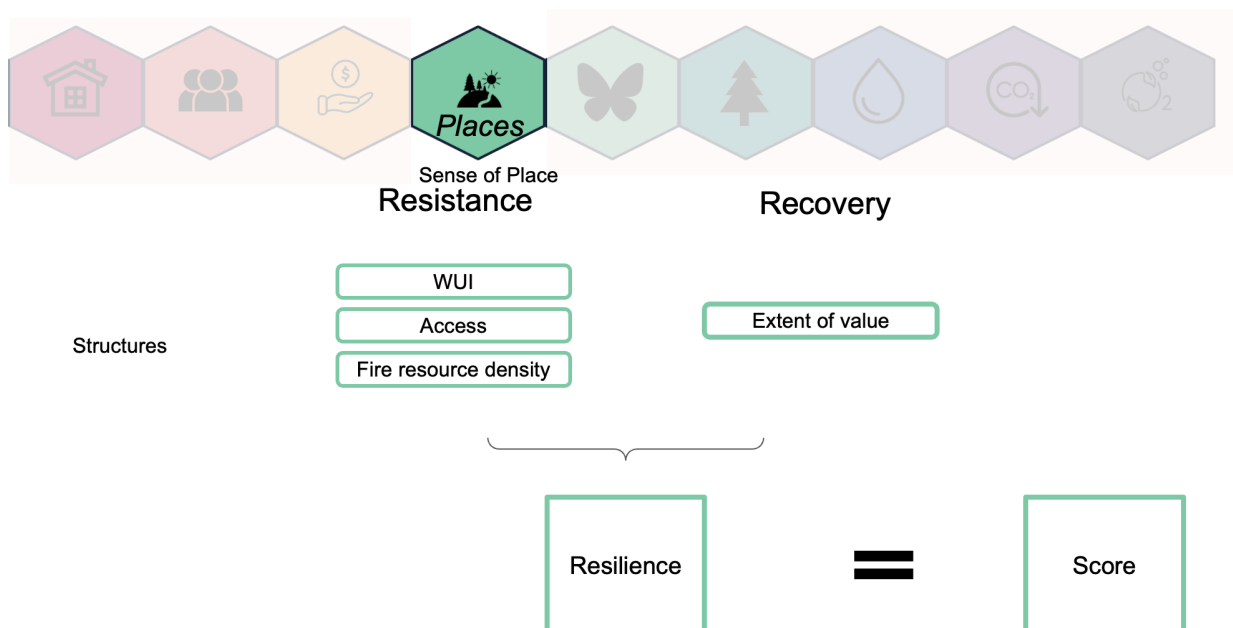


Figure 4. Model for Sense of Place domain, subdomain: Iconic Places subset structures.

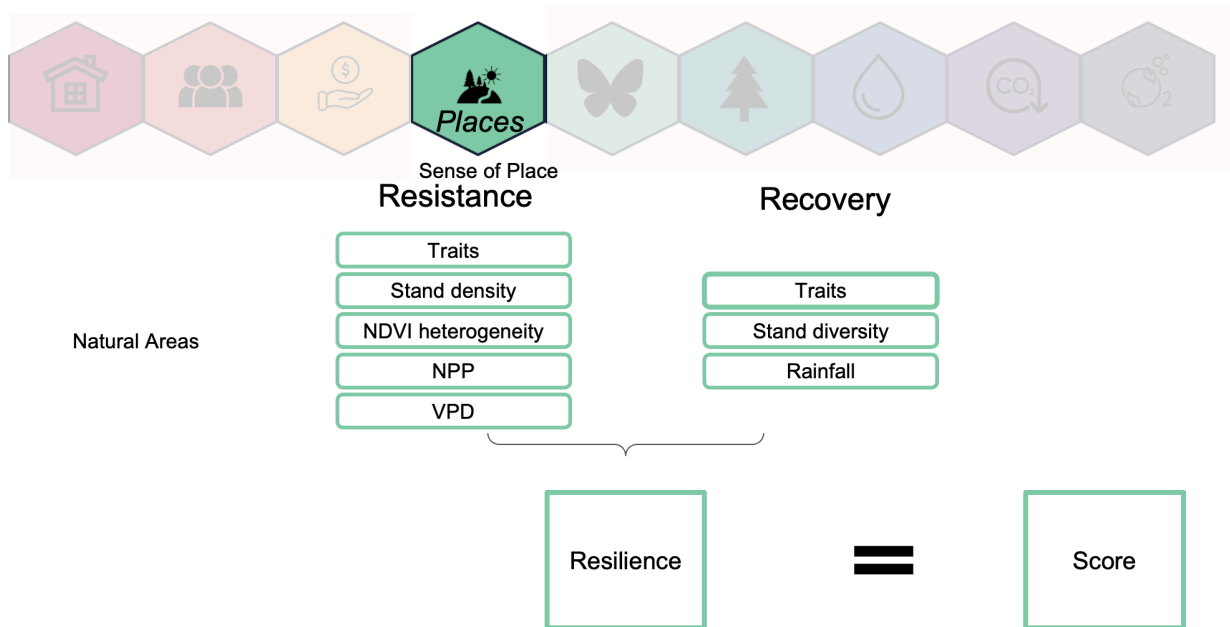


Figure 5. Model for Sense of Place domain, subdomain: Iconic Places subset natural areas.

Presence

We identified iconic places through national registries and national park systems for both the United States and Canada. We have no *a priori* justification for evaluating the iconicness of places because values and importance are highly individual. Therefore, Iconic Places do not have a status score and are simply evaluated where they are. Status and resilience for Iconic Places were evaluated on their footprint, with a buffer of 60 meters around each point geometry.

In the United States, the National Park Service maintains a National Register of Historic Places, which is the federal government's official list of sites, buildings, structures, districts, and objects deemed worthy of preservation for their historical significance or "great artistic value". This register is enabled by the National Historic Preservation Act (NHPA) of 1966, which established both the National Register and the process of adding new places. The Register includes over 1.5 million properties, which are categorized into five types of properties: district, site, structure, building or object. To be eligible for inclusion, a place must meet one of four criteria: contribution to a major event, person, design/construction, or information potential. Inclusion on the list does not automatically qualify a place for special investment in repair or maintenance, however, it does result in a tax credit, eligibility for federal grants, and requires consideration in federal projects.

In Canada, we could not identify an analogous national historic registry. However, both British Columbia and the Yukon territory maintain registries for important historical places ([BC](#) and [Yukon](#), respectively). The British Columbia Register of Historic Places includes officially recognized historic sites in BC as a statutory requirement of the Heritage Conservation Act. This register includes historic places that have been officially recognized for their heritage and cultural values and includes buildings, neighbourhoods, trails, and landscapes that are important to BC's unique identity and heritage. The Register of Yukon Historic Places lists every place designated as historically significant under the Historic Resources Act.

Iconic places also include nationally designated parks in both Canada and the United States. In the United States, national parks are federally protected areas designated by Congress and managed by the National Park Service (NPS), an agency within the Department of the Interior. The NPS was established by the Organic Act of 1916, which mandated the agency to conserve park resources and provide for their enjoyment in a manner that leaves them unimpaired for future generations. National parks are typically designated based on outstanding natural features, scenic value, or ecological significance, as noted in early descriptions (e.g., [Sterling 1931](#)). Among the various land units managed by NPS, only those formally designated as National Parks are included here due to their consistent statutory protections.

In Canada, national parks are established under the Canada National Parks Act, first passed in 1930 and most recently updated in 2000. The system is administered by Parks Canada, a federal agency. The Act mandates that parks be managed to protect their natural heritage for both present and future generations, while also supporting public understanding, appreciation, and enjoyment. Only sites formally designated as National Parks under this Act are considered in this context.

In total we identified 19,834 structures in the US and Canada on the national registries. Of these structures, we identified 218 places that fall outside our study area boundary or are submerged (e.g. lighthouses, shipwrecks), which we excluded from our analysis. For national parks, we identified 84 parks in total in the US and Canada, of which none were excluded.

Resilience

Iconic places include both structural and natural places, which we treat differently in resilience. Broadly, resilience indicators for structures are similar to the Infrastructure domain while resilience indicators for natural places are similar to the Habitats domain.

Resistance

For structures, we include the WUI, egress, and fire resource density indicators from Infrastructure. We do not include the indicators of defensible space or building codes because we posit these are less relevant for historical structures. For example, the value of the structure is not always tied to its integrity (ie, some structures are abandoned or otherwise in disrepair) and in some cases, the landscaping decisions are key to the iconic nature of the structure (e.g. the Ahwahnee).

For natural places, we include all indicators used for Habitats resistance.

Recovery

Extent of significance. For structures, we assess the extent of significance for potential recovery. This extent refers to the level of recognized significance — categorized as local, regional (e.g., state, province, or territory), or national/international. We propose that this extent of significance serves as a useful proxy for the likelihood and availability of resources to support recovery efforts for iconic structures. To convert these categories into an indicator, we assigned a value of zero for locally significant structures, one-half for those with regional significance, such as at the state, provincial, or territorial level, and one for structures with national or international significance. We argue that this extent of significance serves as a useful proxy for the availability of resources to support recovery efforts, since buildings recognized at broader scales are generally more likely to attract attention, funding, and institutional support following damage.

For natural places, we include all indicators used for Habitats recovery.

Iconic Species

OVERVIEW

There are species of special, iconic importance to people that are closely tied to culture and identity. The iconic species subdomain measures the conservation status of iconic species, those plants and animals that have unique importance to humans. We identify these species through their official designation nationally or by the state, province, or territory and/or use on official flags. For Canada, we included species on money; we did not do so for the United States because the U.S. State Quarters program, which ran from 1999-2008 has been discontinued and it is unclear if the contents reflect current values. We compiled a total of 120 species, of which 100 were unique.

MODEL

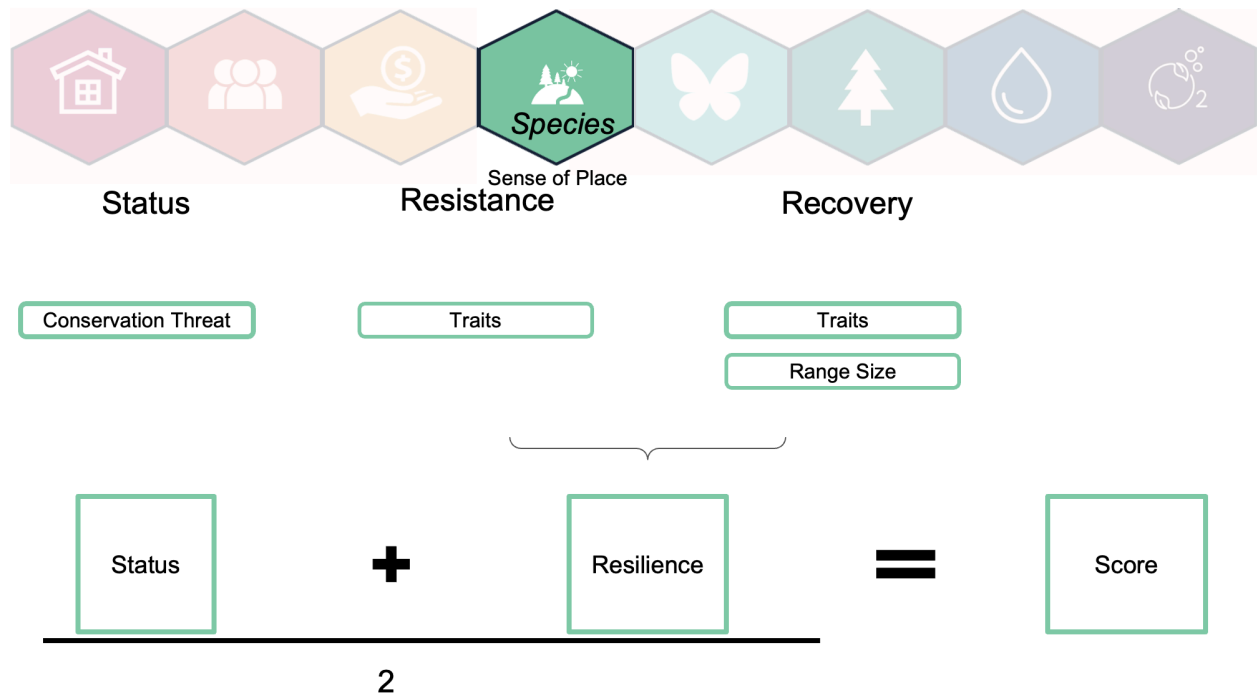


Figure 6. Model for Sense of Place domain, subdomain: Iconic Species

Status

Conservation status. Like for biodiversity, the reference for iconic species is for no species to be at risk of extinction. To evaluate extinction risk we used data on threat status from NatureServe. NatureServe is a nonprofit organization that provides comprehensive conservation status assessments for species and ecosystems, primarily within North America, with growing global coverage. Their threat status rankings are widely used by scientists, conservationists, and policymakers to evaluate the risk of extinction or decline faced by species, as well as the vulnerability of ecological communities.

NatureServe assigns conservation status ranks at multiple geographic scales: global (G-rank), national (N-rank), and subnational (S-rank), which correspond to the species' or ecosystem's condition worldwide, within a particular country, or within a specific state, province, or territory, respectively.

These ranks reflect the level of risk for extinction or elimination. At the highest risk are those ranked as critically imperiled (G1, N1, or S1), indicating an extremely high risk due to very few populations or severe threats. The next level, imperiled (G2, N2, or S2), denotes a high risk resulting from limited populations or significant threats. Vulnerable ranks (G3, N3, or S3) represent a moderate risk where species or ecosystems are rare or threatened. Apparently secure ranks (G4, N4, or S4) indicate species that are uncommon but not rare, with some concern over long-term trends. Finally, secure ranks (G5, N5, or S5) identify species or ecosystems that are common, widespread, and abundant.

Additional ranks include “possibly extinct” (GH, NH, SH), which applies to species not recently located but for which there remains some hope of rediscovery, and “presumed extinct” (GX, NX, SX), used when a species is believed to be extinct with no recent sightings.

NatureServe also uses T-ranks to denote the status of subspecies or varieties. For example, a rank of G5T1 indicates a species that is globally secure overall but has a critically imperiled subspecies. Range ranks, such as G1G3, indicate uncertainty or variability in the status assessment.

These rankings are instrumental in guiding conservation priorities, regulatory decisions, and scientific research. While NatureServe’s system aligns with the International Union for Conservation of Nature (IUCN) Red List, it often provides more detailed, fine-scale information, particularly for North American species. Government agencies, non-governmental organizations, and land managers frequently rely on these ranks to inform habitat protection strategies and recovery planning

We have chosen to exclude the iconic genera Rose, Oak, and Maple from our analysis because, at the country level, no specific species within these genera are identified. Additionally, when a subnational (state or province) NatureServe conservation status score is unavailable for a species, we have decided to use the corresponding national (N) score to fill that gap. Similarly, in cases where the national score is missing, we have opted to infill it with the global (G) score. For iconic birds, NatureServe provides Breeding and Non-Breeding threat status scores, which we averaged.

Based on the range maps available so far, we have identified a few species occurring outside their designated state or province ranges. For example, the horned toad (*Phrynosoma hernandesi*) in Wyoming and the grizzly bear (*Ursus arctos*) in California do not occur in the state that considers them iconic. For these instances, the state-level assessment is recorded as not applicable (NA), reflecting no expectation that these species be reintroduced.

All but six species were in the NatureServe database. These were removed from the subdomain calculation because there is no defensible way of gap filling conservation status.

To quantify extinction risk, we assigned values (which can be interpreted as extinction risk) to every species in a cell based on their threat code (Table 6). We then average the status of all species in a cell and rescaled values to reflect meaningful losses of biodiversity. Specifically, we scaled the lower end of the biodiversity goal to be zero when mean extinction risk was 75% (score = 0.25), a level comparable to the five documented mass extinctions (Barnosky *et al.* 2011), which were catastrophic losses of biodiversity. All species are evaluated and scored within the state or province for which they are iconic. For species that are nationally iconic, those species also get evaluated within their respective states or provinces.

Table 6. Weighting schema for conservation status in NatureServe. B indicates a breeding population, N a non-breeding population, and M a migrant population

Conservation Status Rank	NatureServe Definition	Weight
SX	Presumed Extirpated	0
SH	Possibly Extirpated	0
S1/S1B/S1N	Critically Imperiled	0.2
S2/S2B/S2N	Imperiled	0.4
S4/S4B/S4N	Vulnerable	0.6
S3/S3B/S3N	Apparently Secure	0.8
S5/S5B/S5N	Secure	1
SU/SUN/SUB	Unrankable (lack or conflicting information)	na
SNR/SNRN/SNRB	Unranked (not yet assessed)	na
SNA/SNAN/SNAB	Not Applicable	na

Resilience

We identified two types of resilience indicators, traits and range size. We calculate resistance as the average of all resistance traits for each pixel. For recovery, we first calculate the average trait score for each pixel and then average that mean trait score with the mean range score.

We divided iconic species into three broad taxonomic groups: trees, animals, and plants and mushrooms. We used the same resilience traits for trees as outlined in Natural Habitats and the same resilience traits for animals as outlined in Species. For both groups, we were generally able to query the same databases described in Natural Habitats and Biodiversity respectively. However, iconic species included 15 animals that were not in biodiversity domain, one gastropod and twenty insects, for which we had to manually infill traits from the literature (Supplementary data file). For non-tree plants, we created a new set of resistance and recovery traits.

Resistance

Bark. Like for trees in Habitats, bark helps a plant resist wildfire (Pausas [2014](#)). In this case, we do not have robust information on bark thickness, and the species here are morphologically diverse and range from forbes to woody shrubs. Thus we score bark as present or absent, which received indicator scores of one and zero respectively.

Height. Similar to trees in Habitats, we argue that taller plants are more resistant to wildfire because they have flammable material further off the ground and away from surface fuels. To convert height to an indicator, we performed a comparative rescale, such that the shortest

species received a score of zero and the tallest species received a score of one, intermediate heights were linearly interpolated.

Recovery

Longevity. As for Habitats, we argue longevity is negatively correlated to recovery because longer lived species likely hit key developmental milestones– like age to maturity– more slowly, making recovery both slower and more likely to be interrupted and reset by additional disturbance events. We binned longevity into recovery bins (>1, 1-5, 5-10, 10-20, and 20+ years) which were assigned indicator values of zero to one where progressive bins increased a quarter point in indicator score.

Age to maturity. As for Species, we argue that a shorter age to maturity enables more rapid population growth and faster recovery. This trait is the mean age of first reproduction; if the species is sexually dimorphic, we use female values, as is typical in population ecology. This indicator is rescaled such that the lowest age to first reproduction gets a score of one while the highest age to first reproduction gets a score of zero; all other values are rescaled linearly in between these extremes.

Seed mass. As for Habitats, we suggest smaller seed mass is a life history strategy of bet-hedging against stochastic conditions and events like wildfires, and this strategy of lower investment in more offspring is a recovery strategy for plants.

Resprout. As for Habitats, resprouting, the ability to regenerate after a fire from surviving meristems, is a recovery trait

Range. As for Species, species ranges are mechanistically linked to recovery because parts of the population are more likely to survive in unburned areas and recolonize burned areas. We collected 31 species range map data from Map of Life (MOL [2021](#)). Eight species ranges were obtained through the Map of Life Mammal Diversity database ([2020](#)). For iconic birds, MOL provides the Resident, Breeding, Non-Breeding, and Passage ranges which are combined into one polygon to calculate the range. For eight species, we submitted a request for and received data that are viewable on MOL but not downloadable. In 26 cases where the data were not available in MOL, we used observation points data downloaded through the Global Biodiversity Information Facility ([RGBIF](#)) package accessed from R via *rgbif* ([ref](#)) on June 12, 2025. We removed spatial outliers using a Mahalanobis distance with a 95% Chi-squared threshold. We constructed a concave hull polygon (concavity = 2) around these points to estimate species ranges masked to a North America land boundary. For 10 fish species, we constructed fish ranges by intersecting the RGBIF points to Hydrobasin Level 8 polygons to assign the freshwater ranges because fishes are aquatic and linear interpolation does not make biological sense. 19 tree and flower ranges were obtained from Little [1999](#). One reptile range was obtained from Meiri, Shai; Roll, Uri; Grenyer, Richard et al. ([2017](#)). The total area for each species range across North America was calculated and rescaled between zero and one.

Species

OVERVIEW

People value the existence of a diverse array of species both for their intrinsic qualities and for their contribution to community and ecosystem structure and function. For many, the possibility of species extinction triggers emotional and moral concern, evident in significant financial allocation to species conservation efforts.

This domain assesses the extinction risk of species as well as their resilience to wildfire. Status describes the average extinction risk of species in a region while resilience consists of landscape metric and species-specific traits.

MODEL

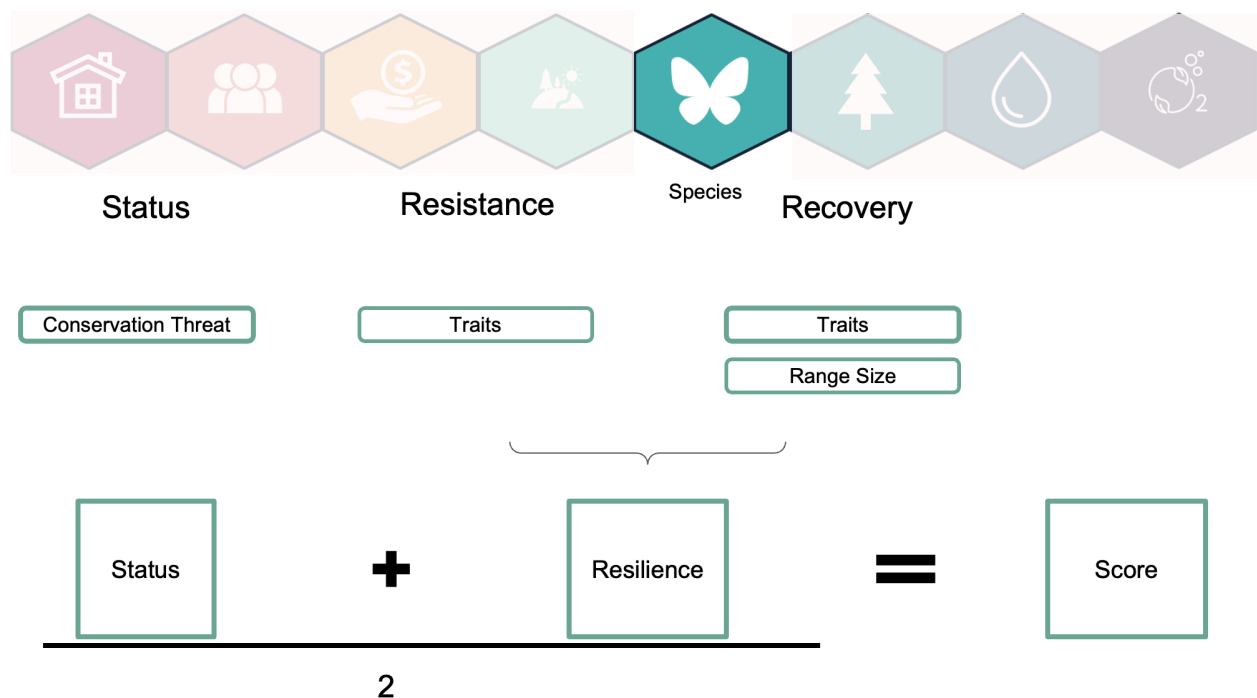


Figure 7. Model for Species domain

Status

Conservation status. The reference for species is for no species to be at risk of extinction. Ideally, status would be a comprehensive evaluation of the conservation status of all life on earth. However, only a small fraction of species have been mapped and assessed.

To evaluate extinction risk, we use data from the IUCN (International Union for the Conservation of Nature) Red List of Threatened Species and BirdLife International, which has been designated by IUCN as the Red List Authority for birds. We used IUCN and BirdLife range maps and threat status evaluation because these data are open access, spatial and policy relevant, and updated frequently. IUCN and BirdLife identify taxonomic groups of interest and evaluate every species in that clade to minimize species selection biases. IUCN range data were downloaded on the website by species grouping while threat statuses were retrieved via the IUCN Red List AIP (v4) in R ([ref](#)). For BirdLife, we contacted the organization for data (DataRequests@birdlife.org). We use IUCN and BirdLife because these organizations provide open access, spatial, policy relevant data layers that are updated frequently, though irregularly.

We focus on animals including freshwater fish, mammals, amphibians, reptiles, and birds, because the data is more comprehensive, enabling comparison across our study region. Plants were excluded from the Species group because the plant species assessed by the IUCN were not representative of the plant species of the study region. We generally excluded marine species. To exclude marine species from the BirdLife species list, we used information from [AVONET](#), a trait database in the Open Trait Network. For the other groups, we included certain marine/terrestrial/freshwater combinations to ensure we included important species like salmon (Table 7). For both data sources, species ranges are represented as vector data and the dataset includes information about threat status, taxonomy, presence, origin, and seasonality.

Table 7. Combinations of species' habitat and justification for inclusion or exclusion; codes are M (marine), T (terrestrial), and F (freshwater)

M	T	F	keep	reason
F	F	F	no	should not exist
T	T	T	no	Significant marine habitat use; implied fire avoidance (2 seals, otter)
T	T	F	no	Significant marine habitat use; implied fire avoidance (5 seals, 2 sea lions, 6 sea turtles, walrus, arctic fox, polar bear, otter)
T	F	F	no	purely marine species
F	T	T	yes	Significant terrestrial use (e.g. amphibians)
F	F	T	yes	Yes, freshwater animals
T	F	T	yes	Yes, anadromous and catadromous species
F	T	F	yes	Yes, terrestrial animals

IUCN and Birdlife attach attributes to each species describing their Presence, Origin, and Seasonality, which are natively encoded as numbers. For the Index, we only evaluated native species, which required filtering of each of these attributes. **Presence** of a species is coded as values 1 through 5. We include extant (1), possibly extant (4), and extinct (5). We exclude probably extant (2) because the code has been discontinued and possibly extant (3) because the probability of occurrence is lower (e.g., because the area is beyond a geographic barrier, or because the area represents a considerable extension beyond areas of known or probable occurrence). We also exclude presence uncertain (6), which represents outliers, unverified accounts, or questionable authenticity. **Origin** of a species is coded as values 1 through 6. We

include native (1) and reintroduced to known historical range (2). We exclude all other categories because they are introduced (3), vagrant (4), origin uncertain (5), and assisted colonization (6). **Seasonality** of a species is coded as values 1 through 5. We include all seasonalities to be comprehensive, including: resident (1), breeding season (2), non-breeding season (3), passage (4), and seasonal occurrence uncertainty) 5. This is the most comprehensive way to include all of the areas a species may use. While passage is of short duration, it could include critical stops in a migration, a known conservation consideration. Similarly, we include seasonal occurrence uncertainty because while time of year for area use is uncertain, the species has been validated as present at some point during the year.

To identify the species in our study area, ranges provided by each dataset were intersected with a vector shapefile of our study area and rasterized. We use an equal area projection (Mollweide) to ensure all cells are equally sized at 1 km to simplify area calculations (see Recovery).

To quantify extinction risk, we assigned values (which can be interpreted as extinction risk) to every species in a cell based on their threat code (Table 8). We then average the status of all species in a cell and rescaled values to reflect meaningful losses of biodiversity. Specifically, we scaled the lower end of the biodiversity goal to be zero when mean extinction risk was 75% (score = 0.25), a level comparable to the five documented mass extinctions (Barnosky *et al.* 2011), which were catastrophic losses of biodiversity (eq. 1). A status score of one means the extinction risk for all species is “Least Concern”.

$$x' = \begin{cases} 0, & \text{if } x < 0.25 \\ \frac{x-0.25}{1-0.25}, & \text{if } x \geq 0.25 \end{cases} \quad (\text{eq. S6})$$

Table 8. Weights for assessment of species status based on IUCN risk categories

Risk Category	IUCN Code	Weight
Extinct	EX	0
Critically Endangered	CR	0.2
Endangered	EN	0.4
Vulnerable	VY	0.6
Near Threatened	NT	0.8
Least Concern	LV	1.0

Resilience

There is one type of resistance indicator and two types of recovery indicators, as for Iconic Species. We calculate resistance as the average of all resistance traits first for each species and then for each pixel. For recovery, we first calculate the average trait score for each species, then for each pixel, and then average that mean trait score with the mean range score.

Traits

We use a trait-based approach to understanding wildfire resilience because functional traits allow for a more mechanistic and predictive understanding of how specific species respond to fire. These approaches focus on functional traits that determine resistance (or survival) and recovery (either through individual return or population demographic processes), making them useful for assessing resilience across diverse taxa.

For this domain, we identify nine functional traits we posit are indicators of resilience, based on the literature (Pausas & Parr, [2018](#); Geary et al. [2020](#); Jones et al. [2023a, b](#)). Some traits are related to both resistance and recovery while others are only mechanistically connected to one dimension. Our nine traits span life history (number of reproductive events, bipartite lifecycle, longevity, reproductive output, age to first reproduction, asexual reproduction) as well as gross morphology (gills, wings, mass).

Life history traits

Number of reproductive events. This trait describes whether an organism reproduces just once (semelparous) or more than once (iteroparous) and is related to the recovery potential of a population. We argue that having multiple reproductive events increases recovery, as opposed to a single event where the entire next generation is dependent on the success of a single event. While we create this binary division, in some ways these strategies can represent a gradient and the timing is more important than the absolute number of events. For example, some dragonflies lay eggs every day for seven days before dying. For the Index, we categorize this a single reproductive event (semelparous) because the events are so tightly constrained in time that this reproductive strategy does not represent a case of distributed risk (Stearns paper). This indicator is binary, such that zero is assigned to semelparous species while 1 is assigned to iteroparous species. (Mahony et al. [2022](#))

Bipartite lifecycle. This trait describes whether the organism has a life cycle with two or more distinct stages that inhabit different environments and the type of environments. We posit this trait is related to resistance because life stages separated in space and time is a bet-hedging strategy against environmental variability, such as wildfire. Examples include transitions between terrestrial and aquatic systems (e.g. amphibians like frogs), transitions from freshwater to marine, or the opposite (anadromous and catadromous species), and host versus free-living stages (e.g. tarantula hawks). This indicator is binary, such that one is assigned to species with a bipartite lifecycle while zero is assigned to all other species. (Mahony et al. [2022](#))

Longevity. We argue longevity is related to recovery such that longer lived organisms are slower to recover simply because population growth is slower. Occasionally, longevity will differ between sexes; in these cases, we used female lifespan, which is typical of population studies. This indicator is rescaled, such that one is assigned to species with the shortest lifespan while zero is assigned to the species with the longest lifespan. All other values are rescaled linearly between these extremes. (Chergui et al. [2020](#), Butt et al. [2022](#))

Reproductive output. This trait is the number of offspring a mature individual produces in a year, which is related to recovery. This indicator is rescaled such that higher annual reproductive rates get a score of one while the lowest annual reproductive rate gets a score of zero; all other values are rescaled linearly in between these extremes. (Bognounou et al. [2021](#), Hu et al. [2016](#), Sitters et al. [2016](#))

Age to first reproduction. This trait is the mean age of first reproduction; if the species is sexually dimorphic, we use female values, as is typical in population ecology. This trait is related to population recovery. This indicator is rescaled such that the lowest age to first reproduction gets a score of one while the highest age to first reproduction gets a score of zero; all other values are rescaled linearly in between these extremes. (Ensbey et al. [2023](#), Butt et al. [2022](#), Mahony et al. [2022](#))

Asexual reproduction. This trait describes whether a species is capable of asexual reproduction, which we argue increases recovery potential by reducing the need to find a reproductive partner. This indicator is binary, such that one is assigned to species with asexual reproduction while zero is assigned to species requiring sexual reproduction. (Butt et al. [2022](#))

Gross morphology traits

Gills. Gills are organs that allow aquatic animals to breathe underwater, and is thus a trait strongly indicative of an aquatic lifestyle, which increases resistance to wildfires. This indicator is binary, such that one is assigned to species with gills while zero is assigned to species without gills.

Wings. Wings have evolved independently across multiple lineages, and in most cases are associated with flight, which we argue is predictive of an individual resistance to a wildfire by fleeing and recovery by returning. This indicator is binary, such that one is assigned to species with wings while zero is assigned to species without wings.

Mass. This trait is the mean adult body mass, which we posit is related to resistance and recovery for the same reasons as wings. This indicator is rescaled such that the largest species gets a score of one while the smallest species gets a score of zero; all other values are rescaled linearly in between these extremes (Lazzari et al. [2022](#), Ensbeey et al. [2023](#), Mahony et al. [2022](#), Butt et al. [2022](#)). We note, however, that body mass does not capture all possible strategies for fire survival; for example, some small burrowing mammals can shelter in place and exhibit higher resistance than medium-sized, less mobile species (e.g., Griffiths and Brook [2014](#)). Our scaling is therefore a

simplification intended to capture general dispersal and recolonization potential, rather than the full range of fire-response strategies.

Trait sourcing, calculations, and gap filling.

When possible, traits were evaluated at the species level. When no information was available at the species level, we used the next closest taxa (e.g. genus, family) and noted when information was obtained at a different taxonomic level (*sensu* Fong et al. 2024, O'Hara et al. 2024). For amphibians, our primary data source was the AmphiBIO database (Oliviera et al. [2017](#)), for reptiles we used the ReptTrait database (Oskyrko et al. [2024](#)) or the Amniote database (Mhyrvold et al. [2015](#)) and for birds and mammals we used the Amniote database (Mhyrvold et al. [2015](#)) for all traits but mass, which came from eBird and Animal Diversity Web respectively.

In some cases, mass was not readily available and we needed to calculate mass from allometric relationships (fish- [Froese et al. 2013](#), amphibians- Santini et al. [2018](#), snakes- Fedman and Meiri [2012](#), lizards- Meiri [2010](#)).

Additionally, we filled in some traits based on assumed relationships between traits. Specifically for fishes, if the time to maturity and longevity were the same, we classified the species as semelparous. Also for fishes, if we could not find information on age to first reproduction, we used generation time as a proxy.

Resistance

The rescaled values from resistance traits (bipartite life cycle, gills, wings, and mass) are averaged to form the final resistance score for each species.

Recovery

Recovery consists of two types of information: species traits and species ranges. For each species, the rescaled and averaged species trait score was averaged with the rescaled range size.

Species traits

The rescaled values from recovery traits (number of reproductive events, longevity, reproductive output, age to first reproduction, asexual reproduction, wings, and mass) were averaged to form the trait score for each species.

Species ranges

Species range is mechanistically linked to recovery because parts of the population are more likely to survive in unburned areas and recolonize burned areas. While there has been substantial discourse in the ecological literature about the merits of a Single Large or Several Small patch for species conservation (SLOSS debate, Tjørve [2010](#)), there is no strong consensus nor are patterns

generalizable across diverse species (refs). Thus, we take the conservative approach of calculating the area occupied by a species and do not account for the spatial distribution of the patch or patches.

We used range map information from IUCN and BirdLife to calculate the total area occupied by a species. For species that are extinct (Presence = 5 extinct, or threat code extinct (EX) or extinct in the wild (EW)), we created a rule where recovery for that species was set to zero. For all other cases, we calculated ranges. There are some key differences for how we filtered IUCN and BirdLife data for recovery compared to status. Notably, we include area outside of our study region for recovery because geopolitical boundaries are not biologically relevant (i.e. Canada, United States, and Mexico). Additionally, we keep all Origin codes, as populations can help with recovery from regions where they are introduced. Further, there are some species that are extinct in our study range, but are extant globally. For these cases, if the species is present somewhere in North America (Presence 1-4), we calculate the range area for that species because this species could return to our study area from intact populations outside our study area. For each species, we calculate range in km² using an equal area projection (EPSG:5070). We then rescaled the indicator between zero and one such that zero is the smallest range and thus has the lowest recovery score and one is the largest range and thus has the highest recovery score; all other values are rescaled linearly between these extremes.

Habitats

OVERVIEW

People care about natural habitats for their intrinsic value as well as for the services and goods they supply. Protecting these habitats is crucial for maintaining global sustainability, which directly impacts human well-being and the survival of many species. Status reflects the current extent of the habitat compared to its historic extent, with the goal of minimizing future loss, as well as percent protected, with the goal of preserving 30% of each ecoregion. To quantify resilience, we combine information on individual species traits, community characteristics, and environmental context.

MODEL

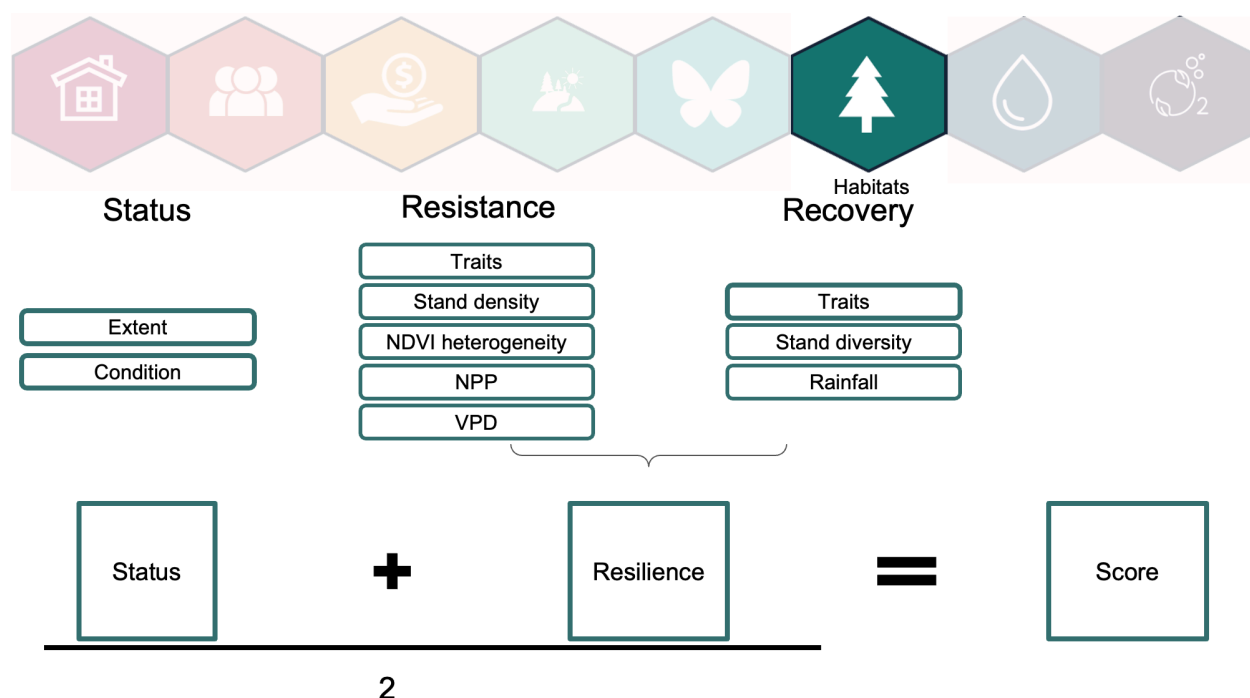


Figure 8. Model for Habitats domain

Status

Status describes the current extent of an ecoregion compared to a historical reference. For all regions, we use 2005 as the reference. Status is calculated as the current extent compared to that historical reference, where loss of an ecoregion to development, either crop or structure, is penalized. Notably, this approach does not account for current or historic logging. Because the metric is bounded between zero and one for use in the composite index, we cannot credit improvements beyond the 2005 baseline (e.g., from restoration or rewilding efforts observed in 2025).

Extent. We calculate extent as current (2023) extent divided by historic extent (2005) for each CEC ecoregion level III in the study area. Current extent is from [10 m Annual Land Use Land Cover \(9-class\) V2](#). The dataset is a machine learning product created by Impact Observatory, Microsoft, and Esri and has an accuracy exceeding 75%. Raw inputs include Sentinel-2 10 m imagery and billions of human-labeled pixels curated by National Geographic for training. Each pixel is classified as one of nine land use and land cover (LULC) types. These maps are an annual composite that result in a representative annual snapshot. Historical extent is derived from GLC_FCS30D, a 30 m land cover dynamics monitoring dataset developed using continuous change detection algorithms applied to Landsat imagery, implemented via Google Earth Engine (Zhang et al. [2024](#)).

Protection. We calculate protection as the proportion of each CEC Level III ecoregion in the study area that is covered by protected areas with robust conservation status. Specifically, we focus on areas categorized as Management Level Ia, Ib, and II in the North American Protected

Areas 2025 dataset, available from the Commission for Environmental Cooperation (CEC). These levels reflect strict biodiversity conservation mandates and long-term protections against conversion. This dataset integrates information from Canada, the U.S., and Mexico and is harmonized across national reporting systems to support trinational conservation goals. We evaluate protection in the context of the global "30 by 30" target, which aims to conserve 30% of lands and waters by 2030 (Eckert et al. [2023](#)). While the dataset reflects the best available spatial information, some areas may be underreported or have ambiguous protection levels, potentially leading to conservative estimates.

Resilience

We conceptualize resilience as a combination of individual, community, and environmental indicators for both resistance and recovery.

Traits

Functional traits offer a powerful approach for understanding ecological patterns, processes, and functions. They facilitate comparisons across diverse ecosystems and geographies (Wright et al. 2004; Díaz et al. 2007; Leichman and Klausmeier 2008; Foden et al. [2013](#); Díaz et al. 2016; Fong et al. 2025). We developed a quantitative trait database for seven fire resilience traits for the dominant species of trees in the western U.S. and evaluated species-level resilience strategies. We then combined this database with spatial data on forest community structure to calculate and map community trait values. We created our species list for the contiguous US using abundance data in TreeMap, a data product that provides 30 m spatial information on forest characteristics (Riley et al. [2021](#)). TreeMap is a product of the Forest Inventory and Analysis (FIA) data and LANDFIRE that models forests across the United States (Riley et al. [2021](#)). For Canadian tree species, we used the Spatialized Canadian National Forest Inventory data product (SCANFI) to determine species list and for spatial characteristics. We focused our analysis on forests of the western states and included tree species that cumulatively comprised 95% of all individuals, resulting in a total of 42 species in the US and 8 species in Canada. The SCANFI dataset included two aggregated categories of tree 'species'- PrcB and PrcC- for broadleaf and conifers respectively. We applied a mean trait value of the other species of broadleaf and conifer to impute trait values for these aggregates. Our trait database includes three traits related to resistance and four related to recovery. Resistance traits include percent bark, height, and self pruning, while recovery traits include longevity, seed mass, resprout, and serotiny.

Individual traits were first rescaled and then combined with community composition data from TreeMap and SCANFI to calculate mean trait value per pixel. We then averaged resistance and recovery traits separately, and that composite value served as the trait indicator value for each pixel for each element of resilience.

SCANFI reports tree composition data using closure percentages while treemap has counts of individual trees and species that make up the spatial data. As a result these two sources must be

rescaled individually and then joined into their aggregate layers. The data are provided in equal area projections that are suited to their designated region to reduce distortion and preserve area and shape. Therefore indicators that use this data, such as tree traits, rescale based on ecoregion in CRS the raw data is provided in and then reprojected into EPSG:5070 to be aggregated into the natural habitats domain scores and eventual index score.

Currently, only trees are included in the trait portion of the analysis. Therefore, we assume that all natural habitat areas that have grasses and shrubs are dependent on the environmental factors only and not the community or individual traits. The exclusion of community and individual traits for grasses and shrubs is due to data limitations at the scale required for this assessment. We recognize that these species are diverse and have traits that impact their resistance and recovery to fire that also affect their larger community.

Resistance

Individual indicators:

Percent bark. Trees that are a higher percent bark (=bark thickness) are less likely to be killed by fire (Keeley & Pausas 2022). Values were rescaled relatively and linearly, such that the species with the highest percent bark was scored as one and the species with the lowest percent bark was scored as a zero, intermediate values were linearly interpolated.

Height. Taller trees may be more resistant to fire by having flammable material further off the ground and away from surface fuels (Stevens et al. [2020](#), Pausas et al. [2004](#)). Values were rescaled relatively and linearly, such that the tallest species was scored as one and the shortest species was scored as a zero, intermediate values were linearly interpolated.

Self pruning. Self pruning is the loss of lower branches, often when light is limiting which increases resistance to crown fire (Pausas [2019](#)). This trait is binary, and the ability to self prune was scored as one while non-self pruning species were scored as zero.

Community indicators:

Stand density. Stand density (the number or cover of trees per area) can influence a forest's resistance to wildfire, though the effects can be context dependent. Generally, higher stand densities are associated with increased mortality because of the increased chance of crown fire or tree-to-tree spread, while lower or moderate densities can contribute to fire resistance under certain ecological conditions.

In high-density stands, the accumulation of live and dead biomass provides abundant fuel for wildfires (e.g., North et al., 2009; 2015). These stands often contain ladder fuels—vegetation that connects surface fuels to the canopy—facilitating the transition from surface fires to more intense crown fires. Forests with lower or moderate stand densities often exhibit characteristics that enhance their resistance to wildfire (Chamberlain et al. [2023](#)). These stands tend to have

discontinuous fuels both vertically and horizontally, which can limit fire spread and reduce fire intensity. Trees in less crowded stands typically experience less competition for resources, leading to greater vigor and greater resistance to fire and pests. Furthermore, such conditions improve the effectiveness of prescribed fire and mechanical thinning as fuel treatments.

Research has shown that many fire-resistant forests in western North America, especially dry pine systems, historically maintained lower stem densities, patchy fuel structures, and were dominated by fire-adapted species such as ponderosa pine (e.g., *North et al.*, [2022](#)). These findings support a restoration approach that emphasizes reducing stand density and reintroducing spatial heterogeneity through active management where appropriate to match historical conditions.

The relationship between stand density and wildfire resistance is not uniform across forest types or regions. In wetter ecosystems, high density may not correspond to increased fire risk, whereas in drier environments, even moderate density can create hazardous conditions if not managed.

We source stand density data TreeMap (count) and SCANFI (canopy coverage). Data were rescaled unimodally by ecoregion to enable comparability across heterogeneous landscapes. We set a density of zero to an indicator value of zero on the lower end while on the upper end we set the 99th percentile to zero to reduce the influence of extreme outliers and all values above the 99th percentile were capped at 0. We set the indicator value of one as the median stand density, which assumes that the central tendency of the dataset represents a reasonable stand density target for an ecoregion, and that this unimodal distribution would effectively penalize pixels with either too high or too low density.

There is no clear corollary for grasses or shrubs, so this indicator pertains to trees only.

NDVI heterogeneity. Research has demonstrated that structural variability can reduce wildfire severity, at least in dry coniferous forests (Koontz et al. (2020)). Fine-scale variability can disrupt fuel continuity and prevent mortality, and that greater variability in forest structure, detected through heterogeneity in the Normalized Difference Vegetation Index (NDVI), increases resilience by reducing mortality. Koontz et al. 2020 found a highly local scale, with best model performance at 90m spatial resolution. However, this model had only modestly higher performance than the model at 270m. Here, we use MODIS 250m NDVI data from [NASA](#), retrieved through the Application for Extracting and Exploring Analysis Ready Samples (AppEEARS) platform and downloaded locally. For each pixel in the study region, we calculated NDVI heterogeneity as the standard deviation of NDVI values within a defined neighborhood. To transform these values into a zero to one indicator, we rescaled standard deviations within each CEC Level III ecoregion. Values were anchored such that a standard deviation of zero (perfect homogeneity) was assigned a score of zero, and the 99th percentile within each ecoregion was assigned a score of one to reduce the influence of extreme outliers. All values above the 99th percentile were capped at one (ceiling), and values below zero were floored at zero, ensuring comparability across heterogeneous landscapes.

NPP. Net Primary Productivity (NPP), the rate at which vegetation accumulates biomass through photosynthesis, is an important ecological variable that can influence wildfire resistance.

Because NPP is closely tied to the amount and continuity of live and dead plant material, it can serve as a proxy for fuel availability in fire-prone ecosystems.

High-NPP regions often accumulate greater biomass, which can lead to increased fuel loads and a higher potential for intense wildfires if other conditions (such as moisture, ignition sources, and continuity) are met. Conversely, landscapes with moderate NPP—such as open woodlands or savannas—often experience frequent, low-intensity surface fires that maintain fuel levels and promote fire-adapted vegetation. In very low-NPP environments, such as arid shrublands and deserts, fires may be rare due to insufficient biomass accumulation (fuels).

We use MODIS 500 m NPP data from USGS, retrieved through the Application for Extracting and Exploring Analysis Ready Samples (AppEEARS) platform and downloaded locally. We rescaled NPP values by ecoregion to enable comparison across diverse and heterogeneous landscapes; for all ecoregions we set the indicator value of one to zero NPP to reflect high resistance; we set the 99th percentile of NPP for each ecoregion to zero to reflect the lowest resistance, intermediate values were linearly interpolated.

Environmental indicators:

VPD. Vapor Pressure Deficit (VPD) is the difference between the amount of moisture the air can hold at saturation and the actual moisture present, and is a key indicator of atmospheric demand for water. High VPD values reflect dry air conditions that draw moisture from vegetation and soil, exacerbating fuel dryness and increasing the likelihood of ignition and fire spread (Clarke et al. [2022](#)) (Table 9, modified from Clarke et al. 2022). Because VPD integrates both temperature and humidity, it serves as an effective proxy for vegetation stress and flammability across ecosystems. In this analysis, we used VPD to represent long-term atmospheric aridity, rescaled within each ecoregion to reflect relative fuel desiccation pressure.

Table 9. Biome and ignition threshold, modified from Clarke et al. 2022.

Biome	VPD _{p=50}
1. Subtropical to Tropical moist broadleaf forests	2.46
2. Tropical and Subtropical dry broadleaf forests	3.05
3. Tropical and Subtropical coniferous forests	2.54
12. Mediterranean forests, woodlands, and scrub	2.31
4. Temperate broadleaf and mixed forests	1.30
5. Temperate coniferous forests	1.54
6. Boreal forests/taiga	1.11

Recent work determined ignition probability as a function of VPD globally for forested WWF ecoregions (Clarke et al. [2022](#)). We selected 50% ignition probability for each ecoregion as our threshold and counted the number of months above the threshold for each pixel. Data was

acquired from TerraClimate and available as monthly averages at 4 km spatial resolution. To convert these monthly counts into an indicator, we set 0 months exceeding the threshold as one (highest resistance) and the 99th percentile for each ecoregion as zero (lowest resistance); intermediate values were linearly interpolated.

The paper detailing VPD thresholds only focuses on forested systems while this study considers all systems. To gapfill, we match non-forests to their closest counterparts (Table 10).

Table 10. Nearest match of non-forested biomes

System	Forest Biome Crosswalk	Justification
<i>Temperate grasslands, savannas, and shrublands</i>	Match to an average of 4 and 5.	Both are temperate, and grasses likely span the range of both
<i>Tundra</i>	Match to 6	While tundra is drier, boreal is best fit, especially for temperature
<i>Deserts and xeric shrublands</i>	Match to 12	While deserts and xeric shrubs are drier, mediterranean is the best fit
<i>Rocks and Ice</i>	none	make all these values 1 (most resilient) because there is no vegetation

Recovery

Individual indicators:

Longevity. We argue longevity is negatively correlated to recovery because longer lived trees likely hit key developmental milestones– like age to maturity– more slowly than shorter lived trees (Shryock et al. [2014](#)) and that the length of this developmental process will make recovery both slower and more likely to be interrupted and reset by additional disturbance events. This indicator was comparatively rescaled, such that the longest lived species received a score of zero and the shortest lived individual received a score of one; intermediate values were linearly interpolated.

Seed mass. We suggest smaller seed mass is a life history strategy of bet-hedging against stochastic conditions and events like wildfires, and this strategy of lower investment in more offspring is a recovery strategy for trees. In short, it reflects a strategy of widespread dispersal with low individual investment. This indicator was comparatively rescaled, such that the species

with the largest seed mass received a score of zero and the species with the smallest seed mass received a score of one; intermediate values were linearly interpolated.

Resprout. Resprouting, the ability to regenerate after a fire from surviving meristems, is a recovery trait. This binary trait was converted into an indicator such that species that can resprout received a score of one and species that can not resprout receive a score of zero.

Serotiny. Serotiny such that fire induces opening of cones is likely a fire adapted recovery trait (Lamont et al. [2020](#)). This binary trait was converted into an indicator such that species that are serotinous received a score of one and species that are not serotinous receive a score of zero.

Community indicators:

Diversity. Higher biodiversity can promote wildfire recovery by providing a form of ecological bet hedging. Diverse communities contain a variety of species with different traits and tolerances, increasing the chances that some species will thrive under the variable and often harsh post-fire conditions. This diversity buffers the ecosystem against uncertainty, allowing recovery processes to proceed despite fluctuating environmental conditions. Essentially, having multiple species spreads the risk and increases the likelihood that at least some will successfully re-establish, supporting faster and more stable ecosystem recovery. Diversity data for the US are sourced from TreeMap (count) while data for Canada are from SCANFI (percent cover). To convert raw data into indicators, we rescaled values within each ecoregion. A value of zero corresponds to zero diversity (i.e., a pixel without trees, and therefore absent from the tree dataset). A value of one corresponds to the 99th percentile of diversity within that ecoregion. All intermediate values were linearly interpolated between these two bounds.

Environmental indicators:

30-year rainfall norm. Wildfire recovery can be influenced by precipitation patterns relative to long-term climate normals. Using 30-year rainfall norms as a baseline, years with near-normal rainfall often provide optimal conditions for vegetation regrowth and ecosystem resilience after fire. When rainfall is close to these historical averages, soil moisture and water availability support seedling establishment and plant growth without the stress of drought or the risk of flooding. In contrast, years with significantly below-average rainfall can limit recovery by imposing water stress, while unusually wet years may also disrupt recovery through soil erosion or waterlogging. Thus, rainfall near the 30-year norm tends to create the most favorable conditions for consistent and sustained post-fire regeneration.

Data were acquired from TerraClimate and available as monthly values, available at 4 km spatial resolution. Historical references were constructed for each pixel to account for microclimates and other drivers of heterogeneity, and spanned 30 years, from 1991-2020. We calculated annual rainfall amounts. To convert precipitation amounts to indicator values, we rescaled each pixel by its historical distribution with a unimodal function, such that an average rainfall amount resulted in a score of one while lower and upper thresholds were pinned to the

1st and 99th percentiles to downweight the effect of outliers. Intermediate values were interpolated linearly with no tolerance around the mean.

Water

OVERVIEW

People need sufficient freshwater that is available at the appropriate time. Wildfires can disrupt the quantity, quality, and timing of surface water.

Here, we evaluate status as the quantity, quality, and timing of surface waters compared to a 30-year historical baseline. Resilience captures interventions by humans that affect water on the landscape and through the tap, including water treatment plants and water management plans. These indicators are only for resistance, there are no recovery indicators because water is a co-benefit of a healthy natural ecosystem, and will recover in tandem with that system.

MODEL

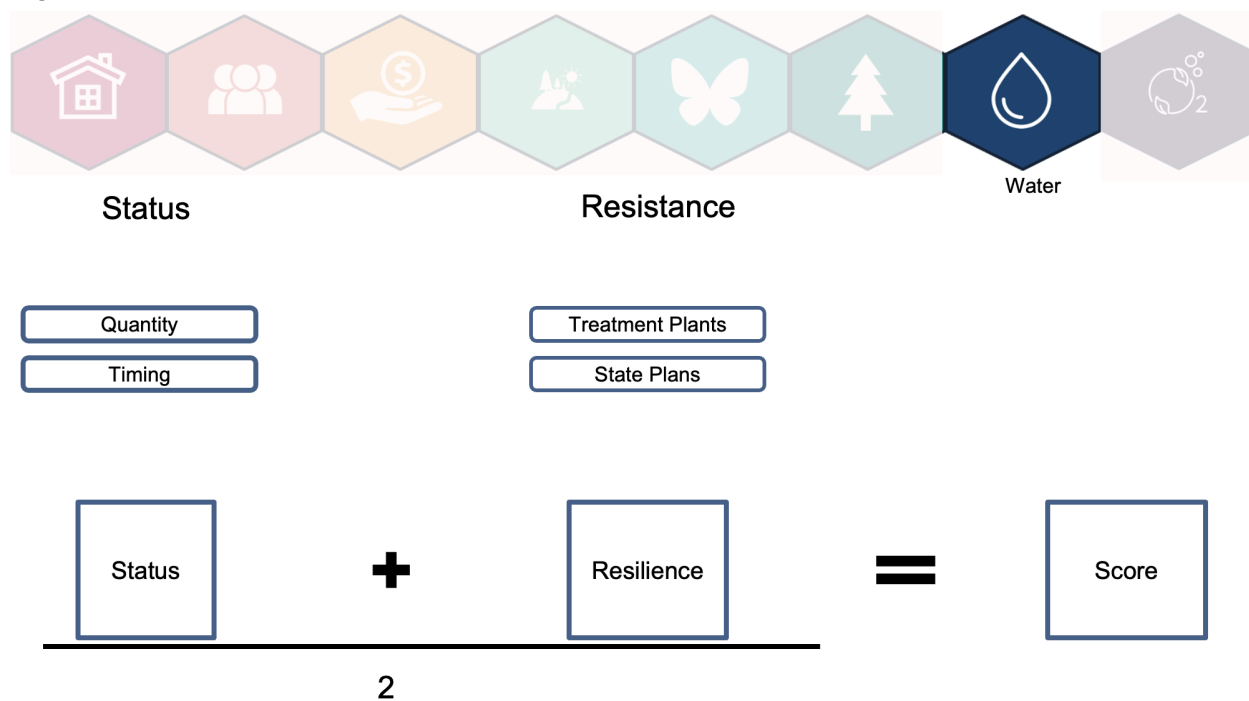


Figure 9. Model for Water domain

Status

Status captures the quantity and timing of water on the landscape. These indicators are compared to a 30 year historical reference to evaluate when indicators are within our outside of

historical norms. Both indicators for status come from longitudinal streamflow data (vol/second), which are dynamic and thus appropriate indicators. Data are from USGS NWIS and hydrometric data from Water survey of Canada, both available as monthly aggregates.

We constructed a historical reference for individual stream gauges, which required us to identify a subset of gauges with robust longitudinal coverage. To do so, we required all gauges to have a minimum of 20 years of complete coverage (all 12 months) within the 1991-2020 reference time period. Additionally, the gauge needed to have complete data for the year of evaluation (in this case, 2024).

Quantity. We assessed total annual discharge by summing monthly streamflow values for each gauge in the evaluation year. To contextualize this value, we constructed a historical distribution of annual totals using a 30-year reference period. This distribution was rescaled to a zero to one range using a unimodal scoring approach: annual totals within ± 1 standard deviation of the historical mean were assigned a score of one, while totals at or beyond the 1st or 99th percentiles were assigned a score of zero. Evaluation-year values were then compared against this distribution and assigned a score accordingly, with intermediate values linearly interpolated to reflect their departure from historical norms.

Timing. We evaluated the timing of streamflow by comparing each month in the evaluation year to its corresponding month in the 30-year historical record. For each gauge, we counted the number of months where discharge fell outside ± 1 standard deviation from the historical monthly mean. This count was then assessed against the historical distribution of monthly outlier frequencies. Gauges with nine or more anomalous months (≥ 99 th percentile) received a score of zero, while gauges with no anomalous months received a score of one. Scores for intermediate values were linearly rescaled to reflect the degree of seasonal departure from historical patterns.

Resilience

Resilience captures the interventions that communities take between water on the landscape and water for human use.

Resistance

Water Treatment Plants. To evaluate a county's capacity to maintain safe drinking water in the event of wildfire-related disruption, we used U.S. Environmental Protection Agency data on Safe Drinking Water Act violations at the facility level. Each treatment plant was linked to the counties it serves, and two indicators were calculated per county: (1) the number of distinct treatment plants providing service, and (2) the total number of violations among those plants. These metrics reflect both redundancy in water treatment infrastructure and historical compliance with water quality standards. Historical Safe Drinking Water Act violations at treatment facilities can indicate vulnerabilities in infrastructure, operational capacity, or resource constraints that may be exacerbated during wildfire events. Facilities with frequent or severe violations are more likely to experience challenges maintaining water quality under the additional stress of wildfire-related

contamination, such as elevated turbidity, increased organic matter, or chemical pollutants from ash and burned materials. Therefore, the total number of violations serves as a useful proxy for potential susceptibility to compromised drinking water quality during wildfire disturbances (e.g., Pennino et al., 2022; Jankowski et al., 2023; Haley et al. [2024](#)). We rescaled each measure to a 0–1 range: counties with more treatment plants received higher scores (most = 1, least = 0), while counties with fewer violations scored higher (least = 1, most = 0). The final indicator was computed by multiplying the two rescaled values, so that counties with multiple high-performing treatment plants received the highest scores—reflecting stronger capacity to sustain safe water access during or after wildfire events. Notably, this indicator is limited to the United States. While the EPA provides comprehensive, plant-level data on violations and service areas, we were unable to identify an equivalent centralized dataset for Canada. Data on water treatment in Canada are decentralized, with responsibility split among provincial and territorial authorities. As a result, we could not construct a comparable indicator for Canadian provinces or territories.

Water Related Hazard Plans. Neither the US or Canada have comprehensive national water related hazard plans. Individual states and provinces are responsible for designing stand-alone plans to reduce hazards and prepare communities for impacts. While a proxy for preparation, we posit having a plan is indicative of resistance capacity.

We used the approach published by Jedd and Smith [2022](#), which evaluated the comprehensiveness of state specific water related hazard plans. We applied the framework developed by those authors to British Columbia and the Yukon territory (Table 11). This work assessed the comprehensiveness of state plans related to water related hazards, including drought, water, climate, and multi-hazard plans. Plans were scored on a series of 9 criteria where more comprehensive plans resulted in higher scores with a maximum possible score of 36.

To convert these scores into an indicator, we rescaled such that a score of zero resulted in an indicator value of zero and a score of 36 resulted in an indicator value of one; intermediate values were linearly interpolated. Thus, this indicator is itself a sub-index.

Table 11. Scoring of water related hazard planning for BC and Yukon Territory. Scoring criteria based on Jedd and Smith [2022](#). For details of scoring, see supplementary data file.

Jurisdiction	Climate score	Water score	Drought score	Multi-Hazard score	Total
BC	6	4	9	4	23
Yukon	4	4	0	2	10

Air

OVERVIEW

Poor air quality has negative health outcomes and is an important dimension of public health and policy, especially in relation to wildfires.

The Air Domain measures the status of an individual's exposure to unhealthy and hazardous air combined with their resilience to poor air quality. We lack information on causal linkages between smoke exposure and health outcomes as well as information on the impact of repeated exposure (Black et al. [2017](#), Gould et al. [2024](#), Lei et al. [2024](#)). Indeed, there is no known safe exposure level–time or concentration– to poor air quality. Thus, resilience in this domain is constrained to indicators related to resistance to negative outcomes from poor air quality exposure.

This domain assesses the current status of air quality as well as resilience of the population to negative health consequences of exposure to poor air quality based on the best available data. The resilience indicators are intended to reflect the potential of negative impacts of poor air quality either as a result of underlying health factors or occupation.

MODEL

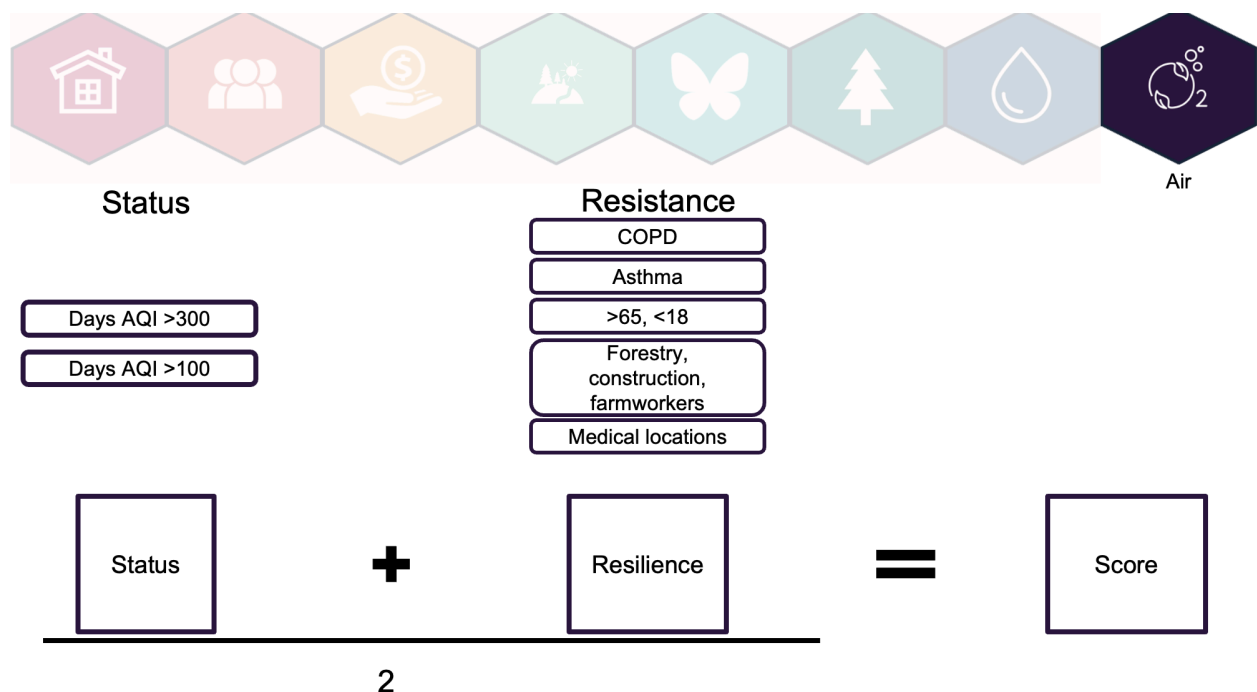


Figure 10. Model for Air domain

Status

Status comprises two indicators: 1) annual maximum number of days an individual is exposed to an AQI of 100 or higher and 2) annual maximum number of days an individual is exposed to an AQI of 300 or higher. The Environmental Protection Agency has developed AQI thresholds above which they posit there are likely increased health risks (Esworthy [2013](#)). AQI greater than 100 indicates a level of unhealthy air exposure for sensitive populations, while AQI greater than 300 reflects acute hazard events for the general population.

Number of days above AQI 100. We used the US AQI as the indicator because it collapses harmful air pollutants (particulate matter 2.5 and 10, ground level ozone, carbon monoxide, sulfur dioxide and nitrogen dioxide) into one number. According to the US EPA, an AQI below 100 is generally thought of as being safe for individuals and still within a healthy range, while values above 100 can cause sensitive groups and the general population to begin experiencing adverse health effects (Plaia and Ruggieri [2011](#)). Notably, Canada uses a different AQI called the Air Quality Health Index (AQHI) which only comprises particulate matter 2.5 and 10, ozone, and nitrogen dioxide. We use a harmonized indicator in our study region for consistency, and choose the US AQI because it is more encompassing of other pollutants.

US data is retrieved through downloading the 2024 Daily AQI by County csv and the Site Listing csv and Monitor Listing csv to attach coordinates to the AQI point location sensors. Canadian daily AQI pollutant data for the six pollutants (ground level ozone, particulate matter 2.5 and 10, carbon monoxide, sulfur dioxide, and nitrogen dioxide) used in the US AQI calculation obtained by emailing dmc@airnowtech.org, who provided the respective csv files for years 2020-2024 for the Canadian bounding box = -141.00, 48.30, -110.00, 69.65 (an approximation of British Columbia, Yukon and Alberta). From this data we identified the max daily AQI at each sensor location for the respective year. The US and Canadian data was then filtered for readings above 100 at each sensor location.

Air quality sensors are not evenly distributed across our study area. To fill in gaps between sensor locations, we applied an inverse distance weighting (IDW) interpolation [approach](#) with a power of five and 10 nearest neighbors. This method, also used by the EPA for AQI mapping (Li et al. [2014](#)), estimates air quality levels in areas without direct sensor coverage by giving greater weight to readings from nearby sensors. IDW operates on the principle that the influence of a sensor diminishes with distance; thus, readings from closer sensors carry more significance in estimating values at any given point, while distant sensors contribute less. Because of this diffusion dimension, we included sensor locations in regions East of our study area (ND, SD, NE, KS, OK, TX, Alberta) for a more accurate representation of air flow.

To standardize and enable meaningful year-over-year comparisons of WWRI indicators, we established threshold limits, setting an upper threshold of 52 and a lower threshold of zero. This means that 52 days with an AQI over 100 acts as a benchmark for consistent exposure and results in the worst possible score. 52 days in a year is equivalent to one day of poor air quality per week (52 days per year), and represents a threshold that we consider a reasonable exposure limit because AQI is meant to indicate an increased likelihood in urgent need for medical care (within a week of exposure). Conversely, the lower threshold of 0 represents an

idealized scenario with no days of poor air quality, serving as a target or best-case reference point. This lower limit reflects the scientific consensus that there is no known safe amount of particulates for human health (Anderson et al. [2012](#)). This upper and lower threshold approach, combined with IDW interpolation, allows us to create comprehensive, spatially-consistent representations of air quality status and trends, supporting equitable assessments of air quality across locations. The indicator is then rescaled between one and zero such that zero is a low status score (i.e. greater number of days of an AQI of 100 or higher) and one is a high status score (i.e. no days above an AQI of 100).

Number of days above AQI 300. As above, we used the US AQI as the indicator. According to the US EPA, an AQI above 300 is considered unhealthy for all individuals. Data are retrieved and processed analogously to above, the key difference is we count days above 300 AQI. We set an upper threshold limit of the 99th percentile (2.83) and the lower threshold was 0 and then rescaled between 0 (greater number of days above AQI 300) and 1 (no days above an AQI of 300).

Resilience

Resilience encompasses the resistance of individuals in a population to poor air quality resulting from wildfire through individual variables broadly describing underlying health conditions and occupational exposure, as well as access to health infrastructure where treatments can increase resistance of affected individuals to lasting negative health outcomes. There are no recovery indicators.

There are five resistance indicators and resistance will be calculated as the average of all the indicators together.

Resistance

Individual indicators of resistance pertain to the vulnerability of an individual to poor air quality due to underlying health concerns or occupational exposure. We also evaluate access to hospitals where treatments can increase resistance of affected individuals to lasting negative health outcomes. We calculate resistance by averaging all indicators.

Prevalence of adult asthma. Asthma is commonly cited as a pre-existing health condition that can be exacerbated by wildfire smoke exposure as well as put individuals more at risk of developing asthma ([Jung et al., 2024](#); [Rappold et al., 2017](#)).

For the US, data are obtained through the Centers for Disease Control (CDC) PLACES: Local Data for Better Health Census Tract 2024 release for adults ≥ 18 . Canadian adult asthma data is available from the Statistics Canada website and is reported at a two year period estimate for the years 2019/2020 for adults ≥ 12 . The spatial resolution of this data is reported at the 12 Health Service Delivery Areas (HSDA) in British Columbia. One value is reported for all of the Yukon Territory.

The data values were assigned to census tract, HSDAs, or territory, as appropriate, and then rescaled between 0 and 1 such that 0 is a low resistance score (i.e. higher rates of adult asthma) and 1 is a high resistance score (i.e. lower rates of adult asthma).

Prevalence of adult chronic pulmonary obstructive disorder (COPD). COPD is commonly cited as a pre-existing health condition that can be exacerbated by wildfire smoke exposure as well as put individuals more at risk of developing COPD later in life ([Jung et al., 2024](#)).

For the US, data is obtained through the CDC PLACES: Local Data for Better Health Census Tract 2024 release for adults ≥ 18 . Canadian percent of adult COPD data is downloaded from the Statistics Canada website and is reported at a two year period estimate for the years 2019/2020 for adults ≥ 30 . The spatial resolution of this data is reported at the 12 health service delivery areas (HSDA) in British Columbia (BC). One value is reported for all of the Yukon Territory.

The data values were assigned to census tract, HSDAs, or territory, as appropriate, and then rescaled between 0 and 1 such that 0 is a low resistance score (i.e. higher rates of adult COPD) and 1 is a high resistance score (i.e. lower rates of adult COPD).

Age. Age is a factor that can increase health risk in wildfire smoke events ([Rappold et al., 2017](#)). Evidence indicates individuals younger than 18 and older than 65 are at higher risk for being affected negatively by smoke.

We calculate the percent of the population that is ≤ 18 and ≥ 65 . US and Canadian vulnerable populations data were obtained through the *tidycensus* and *cansensus* R packages respectively. We used the 2022 American Community Survey 5 year census tract data and the 2021 Canadian census subdivision data. The indicator is then rescaled between 0 and 1 such that 0 is a low resistance score (i.e. many vulnerable individuals) and 1 is a high resistance score (i.e. few vulnerable individuals).

Occupational hazard. There are certain jobs that are inherently outdoors, making it more difficult for people in these jobs to limit exposure to poor air quality. This indicator captures the percent of the total working population that is employed in physical labor jobs that most likely occur outside and put them at risk of breathing unhealthy air in a smoke event and the associated negative health outcomes that may occur due to prolonged smoke exposure. Agriculture and construction jobs are often seen as the most vulnerable jobs ([Jung et al., 2024](#)). We enumerated these workers using the North American Industry Classification System (NAICS) for codes 11 (Agriculture, Forestry, Fishing and Hunting) and 23 (Construction). We calculated the percent of workers in the jobs by dividing by the general working population (ages 15-64).

US and Canadian populations data were obtained through the *tidycensus* and *cansensus* R packages respectively. The Bureau of Labor Statistics Quarterly Census of Employment and Wages (QCEW) survey provides the annual average of monthly employment levels for each NAICS code for each county in the US. We use the most recent full annual data, 2023. Statistics Canada provides the number of workers for each NAICS code for each district in Canada for 2021. Data from Statistics Canada was acquired through the *cansim* R package. The US does not report all farm worker numbers under NAICS code 11, so we supplemented the US NAICS

code 11 worker data with total farm workers and H-2A (temporary agricultural workers) count data from the National Center for Farmworker Health (NCFH). NAICS workers and total farm workers and H-2A workers were summed and the number of workers in these vulnerable occupations were divided by the total working age population (19 - 65) per county or district to calculate the percentage of vulnerable workers in the county and district. The indicator was rescaled between 0 and an upper threshold of the 99th percentile (87.26) as two Washington counties (Grant and Okanogan) had a percentage above 100 meaning there were more workers in vulnerable occupations than census counts on total working age population. The indicator is then rescaled between zero and one such that zero is a low resistance score (i.e. many vulnerable workers) and one is a high resistance score (i.e. few vulnerable workers).

Access to hospitals. Treatments in hospitals can bolster resistance by minimizing negative health consequences. We used the density of hospitals as a proxy to represent an individual's distance to medical infrastructure and care that might offer respirators, ventilators, or other assisted respiratory equipment. A high resistance score for access to hospitals means a region has a high density of hospitals and is a hotspot of access, while areas with few or no nearby hospitals display lower values, highlighting gaps in access.

US and Canadian access to hospitals is determined by obtaining hospital point locations from the Homeland Infrastructure Foundation Level Database (HIFLD), HealthLink BC, and Geomatics Yukon. Hospital locations were identified within our study area and then a kernel density estimate (KDE) was conducted. KDE works by estimating the density of hospitals relative to their proximity to each location within the area. The method involves setting a bandwidth (sigma) or radius around each hospital, which defines the distance over which each hospital can influence surrounding locations. Cells closer to a hospital have higher KDE values, indicating better accessibility, while cells farther away have lower KDE values, marking them as areas with limited access. For the KDE we used the `bw.ppl` function in the `spatstat.explore` R package which uses likelihood cross-validation and is a data-driven bandwidth selector. The indicator is then rescaled between 0 and 1 such that 0 is a low resistance score (i.e. far from a hospital location) and 1 is a high resistance score (i.e. near to a hospital location).